

AIR LAW

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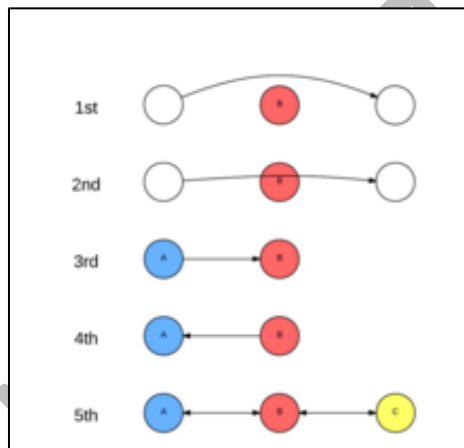
CONVENTION AND FREEDOM

- ICAO convention was signed in Chicago in 1944, 7th Dec
- As per this convention five freedoms of air were declared

1. **FREEDOM 1** - Right to over fly without landing
2. **FREEDOM 2** - Right to land for technical purpose (Non Traffic Purpose)
3. **FREEDOM 3** - Right to put down passenger from home state
4. **FREEDOM 4** - Right to carry passenger for home state
5. **FREEDOM 5** - Right to carry passengers to third state

ICAO World Headquarters: Montreal, Canada

ICAO members are 190 of the United Nations members and the Cook Islands.



CONVENTIONS

1. Tokyo Convention - Offence on Board against penal Law
It is applicable to offences committed on board the aircraft registered in contracting state while in flight, or outside any state. Punishment according to state of registration of aircraft.
2. Rome Convention - Deals with damage caused by foreign aircraft to the third party on the surface (Not Ratified by India)
3. Warsaw Convention - Deals with Responsibility for carriage of passengers, baggage and Cargo.
4. Montreal Convention - Deals with Offences on Board leading to damage of property of aircraft (Unlawful violent act)
5. Chicago Convention - 7th Dec 1944 India is a signatory to Chicago convention.
6. Cape Town Convention: Deals with requirements of aircraft finance and contracts of sale of aircraft equipment (Ratified by India)

7. Hague Convention: Deals with Terrorist hijacking/unlawful seizure of aircraft (Ratified by India)

Objective of ICAO (Chicago Convention 1944) - Is to develop principals and Technique for international air navigation and to foster the planning and development of international air transport.

ICAO also Establishes SARPS – International Standards and Recommended Practices for Contracting states

Submission and Adopting of SARPS - Is done by air Navigation Commission.

- Cabotage Traffic Means Domestic Air Services.
- Open Skies Policy - Free movement of airplanes, crew and passengers not destined for the state in which airplane has landed.
- Any contracting state it denounces the rule of ICAO, it must be informed at least 60 days in advance notice to ICAO
- Wet Lease - When aircraft is given but their crew only flying the plane (Air India plane flown by air India pilots). It will known as Air Indian plane.
- Dry Lease - If aircraft is completely given for use (Air India plane flown by Air force pilots). It will be known as Military/Air Force aircraft.
- Atlantic Ocean - governed by ICAO rules.
- ICAO president is elected every 3 years.

Indian Organisations

Ministry of Civil Aviation

- Responsible for development and regulation of Civil Aviation sector
- Responsible for administration of the Aircraft Act, 1934; Aircraft Rules, 1937 and other legislations related to aviation sector

Directorate General of Civil Aviation (DGCA)

- Responsible for regulation of air transport services to/from/within India
- Enforcement of civil air regulations, air safety, and worthiness standards

Functions of DGCA-

- Registration of civil aircraft, Set standards and grant certificate of Airworthiness
- Licensing of pilots, flight engineers, air traffic controllers
- Granting of Air Operator's certificates
- Investigation into accidents/incidents, Amendment to Aircraft acts, Aircraft rules
- Coordination between civil and military air traffic for use of airspace
- Aircraft Noise, engine emissions, Training programmes

Airports Authority of India (AAI) – founded on 1st April 1995

Functions of AAI

- Design, develop, Operation & Maintenance of international/domestic airports
- Control and Management of Indian airspace
- Expansion and strengthening of Runways, Taxiways, Aprons etc.
- Provision of Visual aids, Communication and Navigation aids, Air Traffic Services
- Provision of aeronautical information services publication of AIP, NOTAMS etc.

ZENITH AVIATION

Annex-I

Personal Licensing

	CPL	CPL + IR	ATPL
Age	Min - 18 yr; Max age - 65 yr	Age limit - same	Min- 21 years, Max -65 yrs
Medical	Validity - 1 year (below 40) 6 months (above 40)	Same	1 yr (below 40), 6 months (above 40)
License	Validity - 5 years	Validity: CPL - 5 years IR - 1 Year	Validity - 2 yrs
Flying	Total flying hours: 200 hrs 100 hrs (SOLO - PIC) + 100 hrs (with instructor) From SOLO:- 20 hr: Cross- Country One-side: (100 NM Min.) 1 X-Country: 300 NM (both sides) 5 hrs Night PIC with min 10 T/O & 10 Landing 10 hrs - IR flight (Min 5 hrs on actual aircraft); 5hr can be on simulator	Total flying hours -200 hrs 100 hrs PIC (Solo ops) 50 hrs: X-Country (100NM) 1 X-Country: 300NM 5 hrs Night PIC with 10 T/O & 10 Landing 40 hrs of IR time (Min 20 hrs on actual aircraft)	Total Hr. - 1500 Hrs 500 Hrs PIC (Not more than 100 hrs as co-pilot) 200 hrs: X-Country 100 hrs Night PIC or co- pilot 100 hrs of IR (Min 50 hrs on actual aircraft) 100 hrs Sim (Not more than 25 hrs on Basic Sim)

- Co-pilot hours will be counted as 50% of flight time for renewal of License otherwise counted as 80% of flight time.
- PPL age: 17 Year
- Consecutive illness of 15 days or hospitalization of 7 day: Authorities must be informed and medical has to be done again.
- Type ratings are done on aircraft which are considered by authority.
- A CPL holder can act as a PIC of an aircraft upto 5700 kg weight and which is entered in aircraft rating of his license.
- Medical validity starts from the day of medical exam.

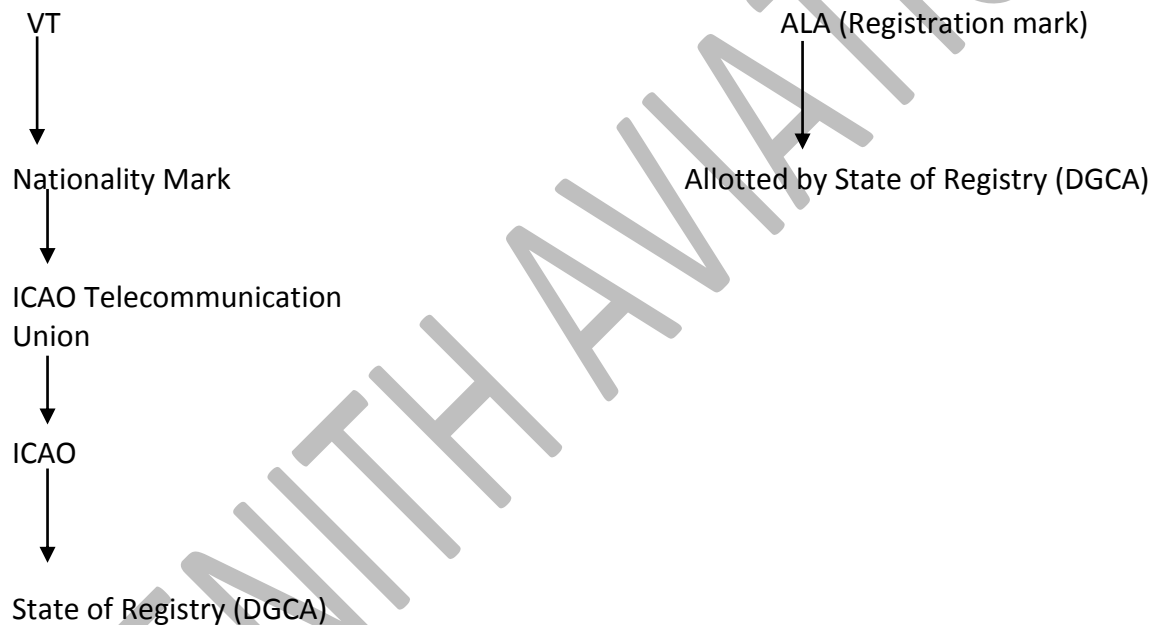
- Instrument Rating holder for a multi-engine aircraft can fly IFR with 1 engine inoperative.
- **Line check:** Line check is to be done within last 12 months (dated from the last day of the month of issue) in any type of aircraft flown.
- To act as a flying pilot (PF), a pilot must carry out 3 take-offs and 3 landings in the last 90 days in an aircraft or in a flight simulator of the same type/class.
- A pilot not holding a valid instrument rating can act as a Commander at night if he has done 1 landing at night in the preceding 90 days as PF in an aircraft.
- To act as a PIC, he must have flown as a co-pilot along the route for 12 months.
- To exercise the privileges of CPL license, pilot must have 5 hours of night flight time including 10 take-offs and 10 landings as pilot in command.
- Pilot can act as PIC if he has made 3 take-offs and 3 landings on the aerodromes along the route within last 3 months.

Annexure - 7

Nationality and Registration Marks



Registration Mark: VT- ALA



- VT-ALA is a five letter combination Registration Mark.
- VT is known as common Mark assigned by Telecommunication Union in ICAO.
- ALA - is assigned by state of Registry (DGCA)
- VT-ALA - this registration is valid until it is cancelled by the authorities or till the aircraft is destroyed in an accident.
- Registration Markings shall be written with no Ornamentation.
- Letter XXX, PAN, TTT shall not to be used as Registration Markings.
- Registration Marks can be alphabets, numeric or alpha-numeric.
- The height of marking on fuselage and underside of wing should be at least 30cm and 50 cm, respectively.
- State of registry (DGCA) is also known as Common Mark Registration Authority

RULES OF AIR-ANNEX 2

Territorial application of the rules of the air –

- The rules of the air shall apply to aircraft bearing the nationality and registration marks of a Contracting State, wherever they may be, to the extent that they do not conflict with the rules published by the State having jurisdiction over the territory over flown.
- If Contracting State hasn't notified ICAO of national differences
- Over airspace and territorial waters of the Contracting state

Responsibility for compliance with the rules of the air –

Rules of the air apply whenever aircraft is in operation, either Flying (IFR & VFR rules apply) or on Ground.

The **pilot-in-command** of an aircraft shall, whether manipulating the controls or not, be responsible for the operation of the aircraft in accordance with the rules of the air, except that the pilot-in-command may depart from these rules in circumstances that render such departure absolutely necessary in the interests of safety.

No aircraft or other object shall be towed by an aircraft, except in accordance with requirements prescribed by the appropriate authority and as indicated by relevant information, advice and/or clearance from the appropriate air traffic services unit.

Parachute descents, other than emergency descents, shall not be made except under conditions prescribed by the appropriate authority and as indicated by relevant information, advice and/or clearance from the appropriate air traffic services unit.

Aerobatic Flight - Manoeuvres intentionally performed by an aircraft involving an abrupt change in its attitude, an abnormal attitude, or an abnormal variation in speed.

Manoeuvring Area - That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons

Movement Area - That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the maneuvering area and the apron

Aerobatic Flight- No Person Shall Fly Aerobatically

- ☐ In the vicinity of an aerodrome at a distance of less than 2 NM from the nearest point of the perimeter of the aerodrome unless being flown higher than 1800m or 6000 feet. (AGL)
- ☐ When flying over city town village or populated area except when written permission by DG.

- ☐ Aerobatics should commence at such a height so it is to be completed at a height not less than 2000 feet above AGL.
- ☐ By a licensed pilot or accompanied by a licensed pilot Instructor.
- ☐ Over aerodrome Minimum Height - 6000 feet AGL.
- ☐ Aerobatics to be conducted with a horizontal separation of 600m from any built up area.

LOW FLYING

- ☐ Aircraft must not be flown lower than 1000ft over a built up area
- ☐ Aircraft must not be flown lower than 500ft in any other area
- ☐ Laterally an aircraft must not be flown closer than 600m to an obstacle/terrain

PROHIBITED AREAS AND RESTRICTED AREAS

Aircraft shall not be flown in a prohibited area, or in a restricted area, the particulars of which have been duly published, except in accordance with the conditions of the restrictions or by permission of the State over whose territory the areas are established.

If a following signal given by ground indicate -

- ☐ "A series of projectiles discharged at intervals of 10 seconds, showing on bursting Red and Green Lights." (First red and Green).
- ☐ Indicates - The Aircraft is flying in the vicinity of a RESTRICTED, PROHIBITED or DANGER area and should take necessary actions.

Necessary actions includes -

- ☐ FIRSTLY - Will not fly further into the Prohibited Area.
- ☐ SECONDLY - Will not descend while still above prohibited Area.
- ☐ THIRDLY - Will ASA he is aware of the fact, give the signal of distress MAYDAY, MAYDAY, MAYDAY by R/T
- ☐ FOURTHLY - Immediately Squawk Distress code "7700" (Column in Aircraft) and land to the nearest aerodrome outside Prohibited Airspace.

Restricted area - An airspace of defined dimensions, above the land areas or territorial waters of a State, within which the flight of aircraft is restricted in accordance with certain specified conditions.

Danger area- An airspace of defined dimensions within which activities dangerous to the flight of aircraft may exist at specified times.

Prohibited area - An airspace of defined dimensions, above the land areas or territorial waters of a State, within which the flight of aircraft is totally prohibited.

Prohibited areas in India

BARODA (Nuclear Reactor)	---- UNLIMITED
BOMBAY (Tower of Silence) (Eagles & Vultures) VA(P)2	1 mile UNLIMITED
MATHURA REFINERY (Sulpohur Smoke) Toxic	10kms UNLIMITED
RASTHTRAPATI BHAWAN VVIP Movements VI(P)-89	----UNLIMITED (1 mile radius)
BHUBANESHWAR TOWER (Nuclear Reactor)	----50000'
KALPAKKAM Tamil Nadu (Nuclear Reactor)	10 km radius upto 10,000 feet height

- The Taj Mahal, Agra, State of Uttar Pradesh, India
- Parliament Building, Prime Minister's residence, and other important centers in New Delhi
- Padmanabhaswamy Temple in Thiruvananthapuram district state of Kerala
- Bhabha Atomic Research Centre
- Sriharikota Space Station in Andhra Pradesh
- The airspace around many Defence and Indian Air Force bases are restricted

FORMATION FLIGHTS

Aircraft shall not be flown in formation except by prearrangement among the pilots-in-command of the aircraft taking part in the flight and, for formation flight in controlled airspace, in accordance with the conditions prescribed by the appropriate ATS authority.

No formation Flights for Civil Aircraft

Right of Way means the right to proceed without alteration of course. The aircraft that has the right of way is required to maintain its heading and speed, and observe the other aircraft whilst the collision risk exists.

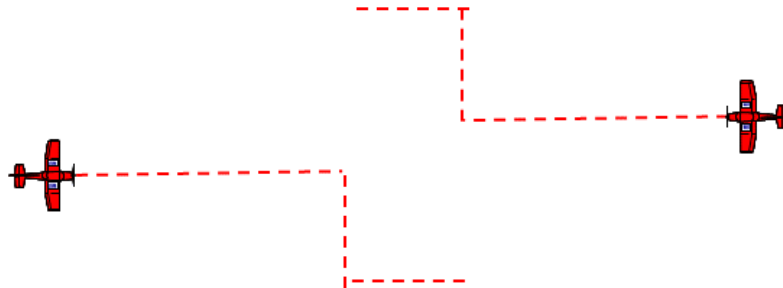
Right of way Approaching head-on-when two aircraft are approaching head-on or approximately so and there is danger of collision, each shall alter its heading to the right.

Aircraft intercepted by another aircraft

- If your aircraft is intercepted by another aircraft follow the instructions of intercepting aircraft and request clarification from ATC
- Once intercepted try communicate on 121.5 MHz, 243.0 MHz or 2182 KHz
- Abrupt 90 degree Maneuver by the intercepting aircraft means you may proceed.
- If communication is not possible on a common language use words in simple like unable, Cannot, Descending, Landing.

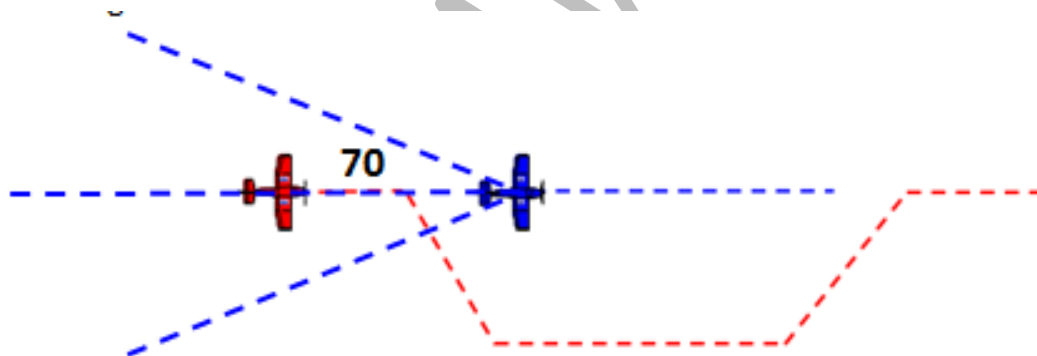
If Hijacked - Squawk 7500 code and start flying 500ft above or below your assigned altitude when below FL290 and 1000ft when flying above FL290.

Head-on Alter Course to Right

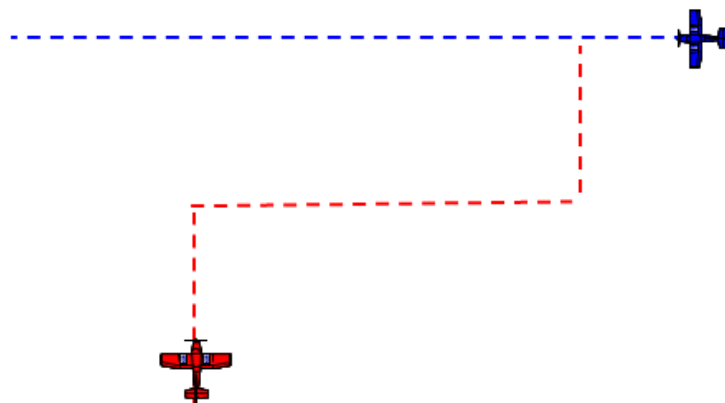


Overtaking - An aircraft that is being overtaken has the right of way and the overtaking aircraft, whether climbing, descending, or in horizontal flights, shall keep out of the way by altering its heading to the right.

An overtaking aircraft is an aircraft that approaches from the rear on a line forming an angles of less than 70°.



Converging - When two aircraft are converging at approximately the same level, the aircraft that has the other on its right shall give way.



GENERAL RULES FOR AIR COLLISION AVOIDANCE

Power driven flying machines must give way to airships, balloons, gliders. Airships must give way to balloons and gliders. Gliders must give way to balloons and power driven flying machines must give way to towing aircrafts.

IMP: The A/C which has other aircraft on its rights shall give way and the aircraft on the right has a right of way.

LANDING

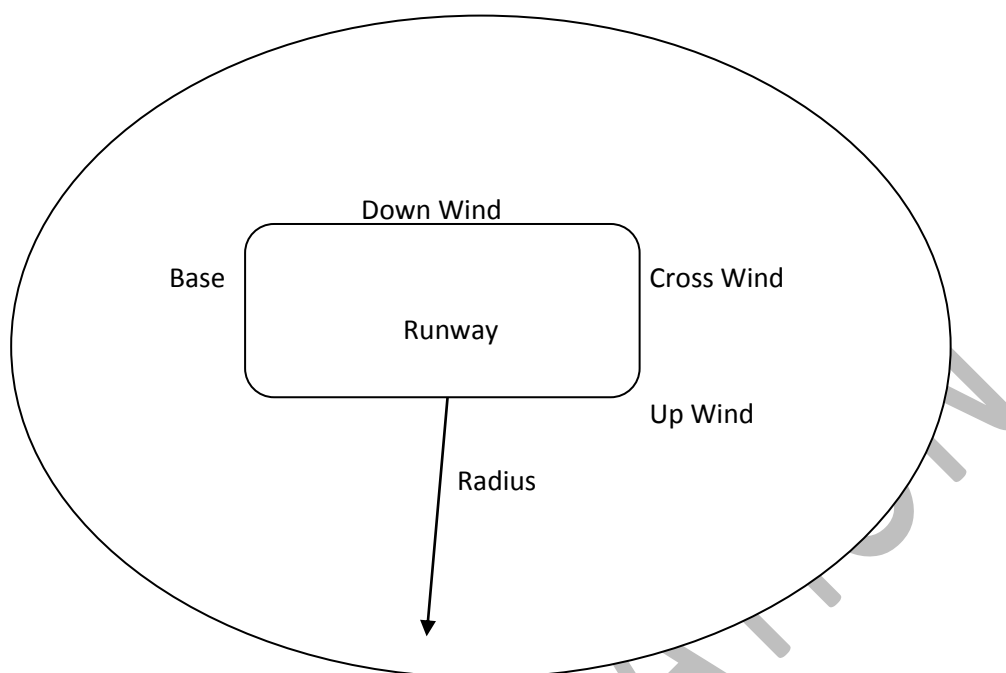
- An aircraft in flight, or operating on the ground or water, shall give way to aircraft landing or in the final stages of an approach to land.
- When two aircraft are approaching for landing, higher one will give way to the lower one, but the latter shall not take advantage of this rule to cut-in in front of another that is on final approach to land, or overtake that aircraft.
- When 2 aircraft are coming in to land, the faster one will give way to the slower one
- If there is risk of collision, the pilot in command (PIC) with right of way must take action to avoid it
- Aircraft with emergency has the right of way
- power-driven aircraft shall give way to aircraft that are seen to be towing other aircraft or objects.

SURFACE MOVEMENT OF AIRCRAFT

In case of danger of collision between two aircraft taxiing on the movement area of an aerodrome the following shall apply:

- ☐ When two aircraft are APPROACHING HEAD ON, or approximately so, each shall stop or where practicable alter its course to the right so as to keep well clear.
- ☐ When two aircraft are on a CONVERGING COURSE, the one which has the other on its right shall give way.
- ☐ An aircraft which is being OVERTAKEN by another aircraft shall have right-of-way and the overtaking shall keep well clear of the other aircraft.

AERODROME TRAFFIC ZONE



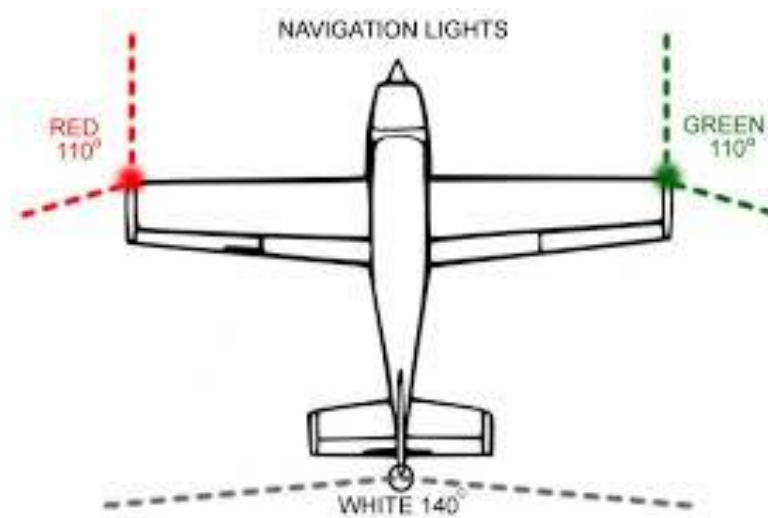
1. Aircraft on final approach for landing - priority over the any other aircraft.
2. Aircraft with emergency - priority over any other Aircraft.
3. Aircraft on runway will have priority over Aircraft on taxiway.
4. Long final 8NM Long.
5. Base leg is reported after completion of turn to the base leg.

LIGHTS TO BE DISPLAYED BY AIRCRAFT

When Lights must be displayed from Sunset to Sunrise (or during any other period prescribed by the appropriate authority), all aircraft in flight or on the movement area of an aerodrome must display:

- ☐ 30 mins after sunset to 30 mins before sunrise considered night in India.
- ☐ Anti-collision lights intended to attract the attention of other aircraft.
- ☐ Navigation lights intended to indicate the relative path of the aircraft to an observer.
No other lights shall be displaced if they are likely to be mistaken for the navigation lights.
- ☐ Unless stationary, and otherwise adequately illuminated, all aircraft on the movement area of an aerodrome shall display lights intended to indicate the extremities of their structure and to attract attention to the aircraft.

- All aircraft operating on the movement area of an aerodrome whose engines are running shall display lights which indicate that fact.



NAVIGATION LIGHTS

The angle of coverage: Parallel to the longitudinal axis to 110° left and other 110° to the right. (Green)

The angle of coverage: Parallel to the longitudinal axis to 70° either side. (White)

ANTI-COLLISION LIGHT

Anti-Collision Lights - red colour flashing rotating 360° beacon (30° above & below horizontal).

All aircraft, in flight or operating on the movement area of an aerodrome, the flight, the light is to be repaired prior to the next flight.

In India, Law is to install it mandatory in aircrafts above 5700 kg of weight.

Failure of Light

- When a pilot is aware that navigation light has failed, ATC is to be informed and the aircraft is to land and have the light repaired before continuing the flight.
- When Anti Collision light fails Aircraft can continue flight and repair it before next flight. Since it is not by law for small aircraft- Less than 5700 Kg.

FLIGHT PLAN RULE - In India, all aircraft going more than 5NM outside or above 3000 ft AGL - must file ICAO flight plan.

IS NEEDED FOR

- ☐ Any flight or portion of a flight requiring an ATC service.
- ☐ Any IFR flight in advisory airspace.
- ☐ Any flight where the appropriate ATC service is required to provide a flight information service (FIS), alerting service, and search and rescue (SAR) service;
- ☐ Where the authority has determined that a FPL should be filed to facilitate coordination between civilian and military authorities;
- ☐ Any flight across international borders.

60 minutes

For a flight to be provided with an ATC service or advisory, the FPL is to be filed at least 60 minutes before departure.

10 minutes

For the filing of a FPL in flight, the FPL is to be filed at least 10 minutes before the aircraft is estimated to reach:

- ☐ The intended point of entry into an area where ATC advisory to be provided; or
- ☐ The point of crossing an airway or airspace advisory route.

DELAYS AFTER FILING

In the event of a delay of 30 minutes in excess of the estimated off-block time for a controlled flight or a delay of one hour for an uncontrolled flight for which a flight plan has been submitted, the flight plan should be amended or a new flight plan submitted and the old flight plan cancelled.

Adherence to the FPL

A pilot operating a controlled flight is required to adhere to the CPL (Current Flight Plan)

Inadvertent Charges

If a controlled flight deviates from the FPL, the following action is to be taken:

- ☐ **Deviation from Track**

Adjust the heading to regain the desired track as soon as practicable.

- ☐ **Variation in TAS**

If Mach No. at cruising level changes (or is expected to change) by ± 0.02 or if TAS changes by $\pm 19\text{km/h}$ or by $\pm 5\%$ or more from that given in the FPL, inform the ATC

☐ **Change in ETA**

If the ETA changes by more than ± 2 mins or more. (Select 3 mins if 2 mins option is not there, earlier it was 3 mins).

Time (ATC clock should be accurate to UTC clocks within 30 seconds).

In all communications, time is to be expressed as Co-ordinated Universal Time (UTC) utilizing the 24 hour clock. A time check is to be obtained before operating a controlled flight.

COMMUNICATIONS RULE

- ☐ A controlled flight is required to maintain two way RTF communications with the controlling ATCU. Air Traffic Control Unit
- ☐ Communication Failure in VMC

If an aircraft is unable to communicate (receive and acknowledge ATC instructions and indicate a state of emergency), in addition to squawking 7600 and maintaining a visual watch for signals, if flying in VMC, **maintain VMC and land at the nearest suitable aerodrome**. ATC is to be informed as soon as possible once the aircraft lands.

VMC flight - Comm. Failure

Light Signal	Aircraft in flight	Aircraft on ground
Steady Red	Do not Land Keep Circling	Stop
Flashing Red	Do not Land Aerodrome Unsafe or closed	Move clear of the Landing Area
Flashing Green	Return to aerodrome & wait landing clearance	Clear to Taxi
Steady Green	Clear to Land	Clear to Take-off
Flashing White	Land on Receipt of Steady Green	Return to starting point of Aerodrome

- ☐ **Irregular switching on and off of Landing Lights indicate that aircraft has emergency & compelled to land. Use Navigation Lights if landing lights are not available.**
- ☐ Red pyrotechnique (Red flare) - Do not withstand previous instructions & do not land for time being.
- ☐ To acknowledge the signals - Aircraft may rock his wings.

If weather is good - Pilot's Choice IFR or VFR

If weather bad - IMC flight only

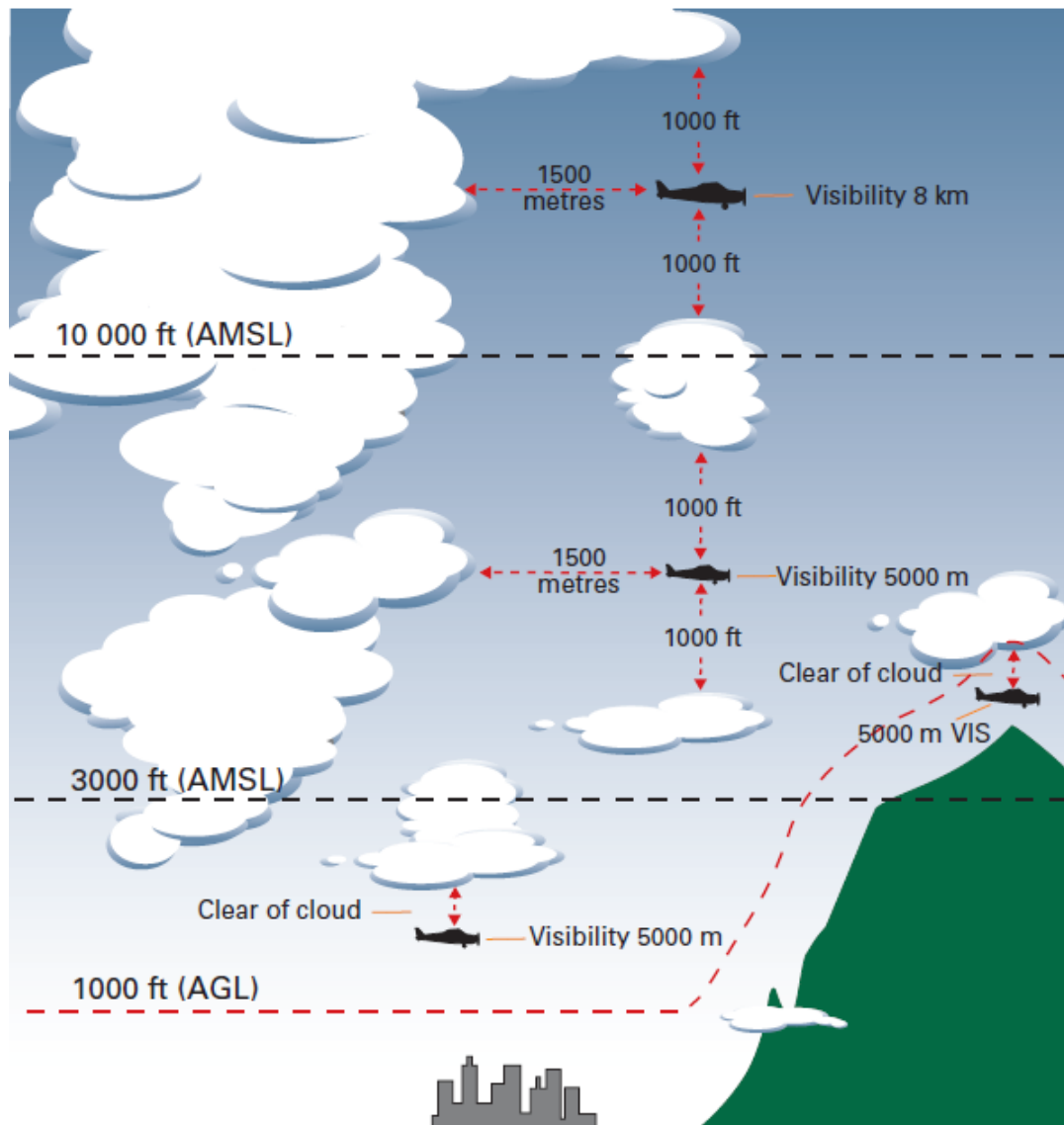
Communication Failure in IMC

- ☐ Shift to VFR if VMC conditions exist
- ☐ Squawk code 7600
- ☐ Maintain the last assigned speed and /or level, for a period of 20 minutes after the failure to report over the last compulsory reporting point;
- ☐ Proceed in accordance with the filled flight plan FPL to the navigation facility serving the destination aerodrome and hold on the facility;
- ☐ Descend from the facility at the last received and acknowledged Expected Approach Time (EAT), or where no EAT has been issued, at the ETA from the FPL; then
- ☐ Fly a normal instrument approach;
- ☐ Land within 30 minutes of the ETA.

Estimated time of arrival (ETA) - For IFR flights, the time at which it is estimated that the aircraft will arrive over that designated point, defined by reference to navigation aids, from which it is intended that an instrument approach procedure will be commenced.

Expected approach time (EAT) - The time at which ATC expects that an arriving aircraft, following a delay, will leave the holding point to complete its approach for a landing. A revised EAT is transmitted when it differs from previously assigned by 5 mins or more.

RULES OF THE AIR FLIGHT RULES - VFR AND IFR



This weather or better than this weather is called VMC.

Instrument meteorological conditions - Meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling, less than the minima specified for visual meteorological conditions.

Visual meteorological conditions - Meteorological conditions expressed in terms of visibility, distance from cloud, and ceiling, equal to or better than specified minima.

Special VFR flight - A VFR flight cleared by air traffic control to operate within a control zone in meteorological conditions below VMC

Meteorological Visibility - Visibility for aeronautical purposes is the greater of:

- The greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;

- The greatest distance at which lights in the vicinity of 1000 candles can be seen and identified against an unlit background.

Flight Visibility – The forward visibility from the cockpit of an aircraft in flight.

Ceiling - The height above the ground or water of the base of the lowest layer of cloud below 6000 meters (20000 feet) covering more than half the sky. (Ceiling- The height of clouds from ground).

Cloud Base – The height of the **lowest layer of clouds** from the ground.

Class of Airspace	Vertical Limits	Visibility	Distance from Cloud
A, B, C D & E and F & G (Above 3000ft AMSL or above 1000ft above terrain, whichever is higher)	At and above FL 100	8Km	1000ft vertical, and 1500m horizontal from cloud
	Below FL 100	5Km	
F and G At and below 3000ft AMSL or 1000ft above/terrain whichever is the higher		5Km	Clear of cloud and in sight of the surface

TAKE-OFF AND LANDING

VFR flights cannot take-off or land at an aerodrome in a CTR, or enter an aerodrome traffic zone or traffic pattern when:

- ❖ The ceiling is less than 1500 ft
- ❖ The visibility is less than 5 Km

NIGHT - NO VFR AT NIGHT

MINIMUM HEIGHTS

A VFR flight shall not be flown:

- ☐ Over the congested areas of cities, towns or settlements, or over an open air assembly of persons, at a height less than 300 m (1000 ft) above the highest-obstacle within a radius of 600 m from the aircraft.

- ☐ Elsewhere, at a height less than 150 m (500 ft) above the ground or water.

An aircraft must not conduct a VFR flight at a height of, or less than, 2000 feet above ground or water if:

- (a) the pilot is not able to navigate by reference to the ground or water; and
- (b) No permission from DGCA that the flight may be conducted at a height of 2000 feet or less.

MINIMUM HEIGHT RULE (IMC)

- Over high terrain or in mountainous areas, at a level which is at least 600 m (2000 ft) above the highest obstacle located within 8 km of the estimated position of the aircraft;
- Elsewhere than as specified in a), at a level which is at least 300 m (1000 ft) above the highest obstacle located within 8 km of the estimated position of the aircraft.

IFR is flown whenever weather is below VMC and when Pilot wishes to fly IFR but with proper navigational equipment's on board

Weather Deterioration below VMC

If it becomes evident that a controlled VFR flight will not remain in VMC, the pilot is to:

- ☐ Request an amended clearance to continue to the destination aerodrome by another route remaining in VMC;
- ☐ Land at the nearest useable aerodrome; or
- ☐ If operating in a CTR, request a SVFR clearance; or
- ☐ File an IFR FPL.

SPECIAL VFR (SVFR)

SVFR is defined as a clearance to fly within a CTR zone in conditions less than VMC in which the pilot remains clear of cloud and in visual contact with the ground.

Provision of SVFR

Where a pilot cannot, or has good reason not to, comply with IFR in a CTR he/she may request a special VFR (SVFR) clearance to:

- Enter a CTR to land at a aerodrome within the CTR;
- Take off from an aerodrome within a CTR and depart from the CTR; or
- To fly between aerodromes within a CTR.

Take-off Conditions

According to ICAO a SVFR flight may take off from an aerodrome in a CTR providing the ground visibility is not less than 1500 m

Reduced Vertical Separation Minima (RVSM)

Semi – Circular Rule

- Flights are to maintain Vertical separation of 1000ft between flight levels
- Flying towards East – Flights on 0 to 179 Magnetic track
- Flying towards West – Flights on 180 to 359 Magnetic track

IFR

- Flights towards East maintain Odd FL;
- Flights towards West maintain Even FL

VFR

- Flights towards East maintain Odd FL + 500ft;
- Flights towards West maintain Even FL + 500ft

The 1000ft min separation between flights follows only upto FL290 because after this height the density decreases and so does the accuracy of the flight instruments.

Above FL290, Minimum separation between flights is increased to 2000ft.

Min. separation below same track aircraft above FL290 is 4000 ft.

The desired cruising levels for turbo-jet aircraft are adjacent to the tropopause (typically FL350- 370). In the upper airspace these levels quickly become occupied and congestion arises.

To overcome this, a system of reduced vertical separation is applied where the 1000ft separation between FLs is maintained up to continuation of the reduced minima. This immediately doubles the available FLs between FL290 and the limit of the application, FL410.

This standard is known as Reduced Vertical Separation Minima (RVSM)

REQUIREMENTS

To fly in airspace where RVSM is applied, an aircraft must be equipped with -

- ☐ Two independent altitude measuring systems;
- ☐ An altitude alerting system (activated by deviation from the selected altitude);

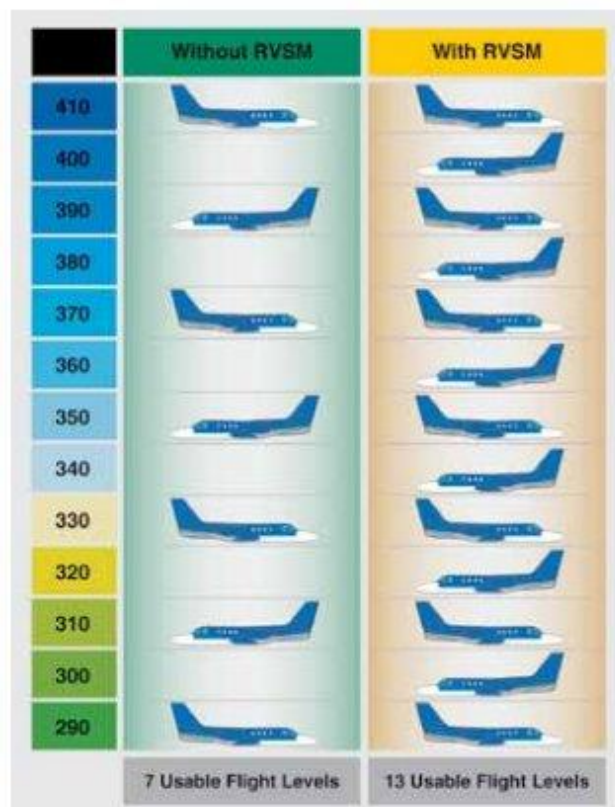
- ☐ An automatic altitude control system (height lock),
- ☐ An SSR system with altitude reporting (mode C) connected to the system used for the automatic altitude control system.
- ☐ Additionally, the operator must be approved for RVSM operations.

Reduced vertical separation minima or minimum (RVSM) is the reduction of the standard vertical separation required between aircraft flying between FL290 (29,000 Ft) inclusive, from 2,000 feet to 1,000 feet. This therefore increases the number of aircraft that can safely fly in a particular volume of airspace.

CVSM - Conventional Vertical Separation Minima

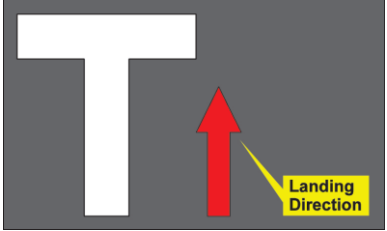
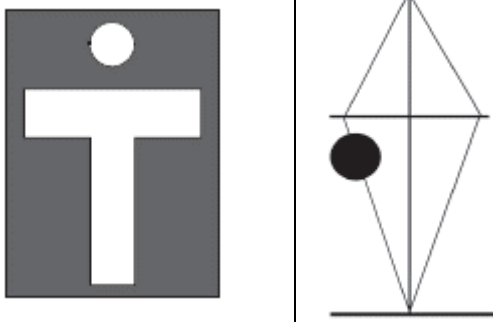
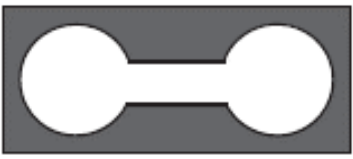
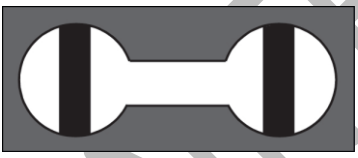
1. RVSM airspace starts from FL290 to FL410
2. Minimum separation above FL290 is 2000 ft
3. Min. separation below same track aircraft above FL290 is 4000 ft
4. Min. separation above FL290 in RVSM airspace is 1000 ft
5. Min. separation in RVSM above FL290 for same track is 2000 ft
6. Min. separation below FL290 is 1000 ft opposite traffic and 2000 ft (same side)

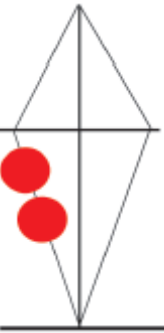
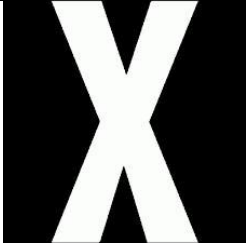
Unless mentioned, Questions are to be solved as per CVSM.



Visual Aids

Signals in the Signals Area

Symbol	Name	Meaning
	Landing T (White or Orange)	Take off & Landing has to be in the direction of the T
	T with a Disc Or Black ball hanging from a mast	Take off & Landing direction may not be the same.
	White dumb - bell	Take off, Landing & Taxi confined to paved surface only (soil on land is not strong)
	White dumb-bell with black perpendicular bars	Take -off & landing confined to paved surface only but taxi (& parking) need not to be confined to hard surface
	Single yellow diagonal on a red background	Maneuvering area is poor-exercise special care when landing
	Two yellow diagonals on a red background	Aerodrome is unsafe for the movement of the aircraft & landing is prohibited

		Red & Yellow tripped right Arrow Or Green flag hanging from a mast	After take-off, right hand circuit pattern is in use. (means you have to turn right after take-off)
		Double White Cross Or Two red balls hanging from a mast	Glider flying is in progress (so that you must be ready to give way)
		Black Letter C on a yellow background	ATC reporting office (to submit the flight plan & report after flying) where visiting pilots must report
		White cross on black background	Runway closed
		Yellow cross	Taxiway closed
		Crossed Hatched Red-Yellow Flag	Taxi only in accordance with ATC instructions

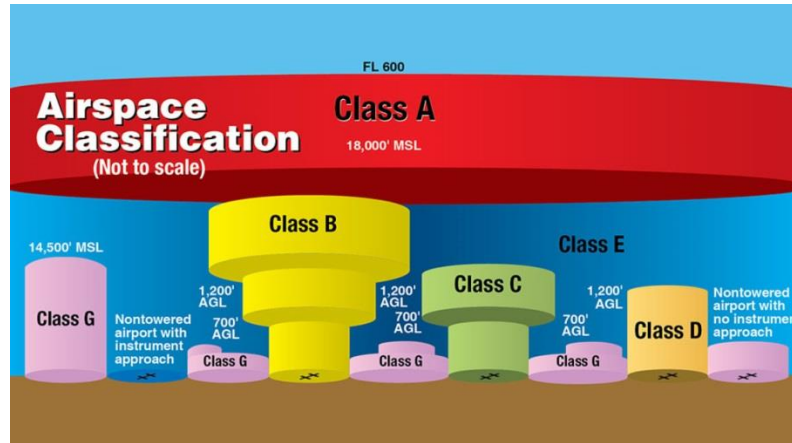
Flight Deck Signals

Signal	Meaning	How to do it
	Brakes engaged.	Raise arm and hand with fingers extended horizontally in front of face, then clench fist.
	Brakes released.	Raise arm with fist clenched horizontally in front of face, and then extend fingers.
	Insert chocks.	Arms extended palms facing outwards, move hands inwards to cross in front of face.
	Ready to start engine indicated.	Raise the number of fingers on one hand indicating the number of the engine to be started. For this purpose the aircraft engines shall be numbered as follows, No. 1 engine shall be the port outer engine, No. 2, the port inner engine, No. 3, the starboard inner engine and No. 4, the starboard outer engine.
	Remove chocks.	Hands crossed in front of face, palms facing outwards, move arms outwards.

AIR TRAFFIC SERVICES

(Annexure – 11)

Airspace Classification

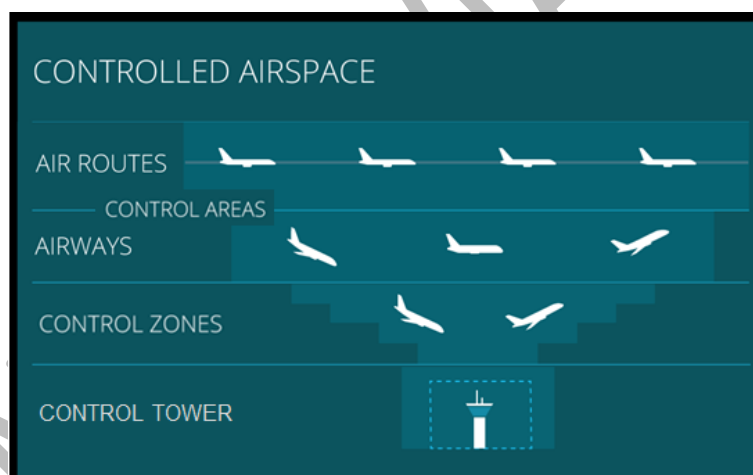
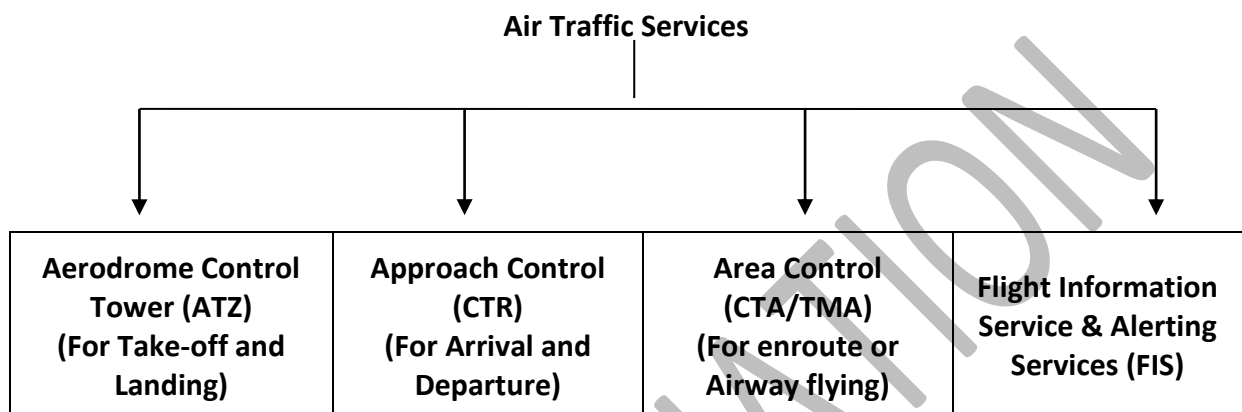


Air Traffic Control Service: ATC services are provided for the purpose of:

- (a) Preventing collisions:
 - (i) Between aircraft
 - (ii) On the manoeuvring area between aircraft and obstructions; and
- (b) Expediting and maintaining an orderly flow of air traffic.

- **Air Traffic Control Unit** - A generic term meaning variously, area control centre, approach control office or aerodrome control tower and FIS & Alerting Services.
- **Air Traffic Services Airspace** - Airspace of defined dimensions alphabetically designated, within which specific types of flights may operate and **for which air traffic services and rules of operation are specified.**
- **Controlled Airspace** - Airspace of defined dimensions within which **air traffic control service is provided to IFR flights and to VFR flights in accordance with the airspace classification.** Note. - Controlled airspace is a generic term which covers ATS airspace classes A, B, C, D and E.
- **Controlled Aerodrome** - An aerodrome at which air traffic control service is provided to aerodrome traffic.
- **Alerting Service** - A service provided to notify appropriate organizations regarding aircraft in need of search and rescue aid, and to assist such organizations as required, provided by FIS (Flight Information Services).
- **Air Traffic Advisory Service** - A service provided within advisory airspace to ensure separation, as far as practical, between aircraft which are operating on IFR flight plans. (Provided only in F type of airspaces)
- **Approach Control Service** - Air traffic control service **for arriving or departing controlled flights** those who have filled flight plans.

- **Control Zone (CTR)** - A controlled airspace extending upwards from the surface of the earth to a specified upper limit. CTRs are defined to include all the airspace, used for IFR flights arriving to and departing from aerodromes. The lateral limit of a CTR should extend at least 9.3 Km (5 nm) from the centre of the aerodrome, or aerodromes.
- **Control Area (CTA/TMA)** - A controlled airspace extending upwards from a specified limit above the earth. The lower limit of a CTA is not less than 200 m (700 ft) above MSL or the (AGL) ground (whichever is higher - some aerodromes can be below sea level).



Airspace Classes

Class A: Only IFR flights are permitted; all flights are subjected to ATC and are separated from each other.

Class B: IFR and VFR flights are permitted; all flights are subjected to ATC and are separated from each other.

Class C: IFR and VFR flights are permitted; all flights are subjected to ATC, and IFR flights are separated from other IFR flights and from VFR flights. VFR flights are separated from IFR flights and receive information about other VFR flights.

Class D: IFR and VFR flights are permitted; all flights are subjected to ATC. IFR flights are separated from other IFR flights and receive traffic information concerning VFR flights. VFR flights receive traffic information for all other flights.

Class E: IFR and VFR are permitted; IFR flights are subjected to ATC. IFR flights are separated from other IFR flights. All flights receive traffic information as far as is practicable. Class E is not used for CTRs.

Class F: IFR and VFR flights are permitted. All participating IFR flights receive **ATC advisory service** and all flights receive flight information service if requested.

Class G: IFR and VFR flights are permitted and receive flight Information service if requested.

Class	Rules	Service	ATC Clearance	Separation	FIS	Speed Limit	Radio	SVFR	Use
A (1)	IFR	ATC	Required	IFR-IFR		No	Yes	Yes	Airway; CTA; CTR; ATZ
B (4)	IFR	ATC	Required	IFR-IFR IFR-VFR VFR-IFR VFR-VFR		No	Yes	Yes	Airway; CTA; CTR; ATZ
	VFR	ATC							
C (3)	IFR	ATC ATC	Required	IFR-IFR IFR-VFR VFR-IFR	VFR-VFR	No	Yes	Yes	Airway; CTA; CTR; ATZ
	VFR	ATC FIS				Yes			
D (1)	IFR	ATC ATC	Required	IFR-IFR	IFR-VFR	Yes	Yes	Yes	Airway; CTA; CTR; ATZ
	VFR	ATC FIS		None	VFR-IFR VFR-VFR				
E (1) (Practicle)	IFR	ATC FIS	Required	IFR-IFR	(As far as practicle) IFR-VFR	Yes	Yes	No	Airway; CTA
	VFR	FIS FIS	None	None	VFR-IFR VFR-VFR		No		
F (1) (Advisory)	IFR	ADV FIS	None	IFR-IFR	All	Yes	Yes	No	Advisory Routers
	VFR	FIS	None	None			No		
G (0)	IFR	FIS	None	None	All	Yes	Yes	No	FIS Routes Open FIR
	VFR						No		

- **C To G airspaces includes FIS Service**
- **Speed: Max 250 knots IAS below FL 100 (From airspace C (VFR) till G)**
- **Airspace A to D is generally considered Controlled AD and E to G are considered Uncontrolled AD.**
- Aircraft transmitting from one FIR (Flight Information Region) to another FIR should inform ATS unit atleast 10 mins before entry to get FIS.
- Domestic flight carrying passengers should fly atleast 15 NM away from international border.

ATC Service shall be provided:

To all IFR flights in A,B,C, D
 To all VFR flights in B,C,D
 To all special VFR flights
 To all traffic in controlled airspace

Vertical Separation is provided:

Between all flights in A & B airspaces
 Between IFR flights in C,D & E
 Between IFR flights and VFR flights in C
 Between IFR flights and special VFR flight
 Between special VFR flights

ATS Route Designator:

A, B, G, R - International Non-RNAV Routes
 L, N, M, P - International RNAV Routes
 Q, T, Y, Z – Domestic RNAV Routes
 H, J, V, W – Domestic Non-RNAV Routes
 *International routes are also known as Regional network routes.

Non-RNAV Routes – Routes which are based on ground navigational aid only.

RNAV Routes – Routes which are based on ground and satellite based navigational aids.

Routes in India:

- W - Domestic route
- U - Upper Airspace
- S - Supersonic route
- K - Low level routes (helicopters)

Airspaces in India

- **Class D:** Airspace in control areas (CTA), terminal areas (TMA), control zones (CTR) and aerodromes traffic zones (ATZ)
- **Class E:** Airspace in ATS routes outside CTAs/TMAs/CTRs/ATZs; Except ATS routes segmented listed under class F.

- **Class F:** Airspace outside CTAs/TMAs/CTRs/ATZs.

ATIS (Automatic Terminal Information Service)

- ATIS is a voice broadcast of current weather of the aerodrome **within 5 NM radius and 5000 ft (1500 m) height.**
- ATIS broadcast can be on a **discreet VHF frequency or on a voice channel of VOR, but can never be on the voice channel of ILS.**
- ATIS broadcast is **updated upon receipt of any official weather regardless of content change from MET (Meteorological) department or immediately after a significant weather change.**
- Preparation and dissimulation of ATIS is a **responsibility of Air Traffic Services.**
- It is a voice broadcast of essential information in a high activity terminal area.
- ATIS broadcast message, when practicable, should not exceed 30 secs.

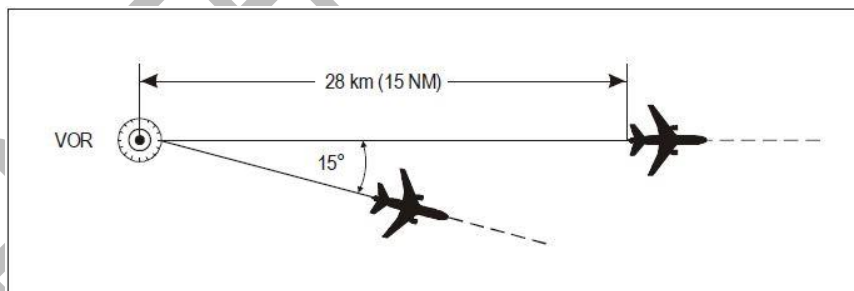
Separation Methods and Minima

- **Horizontal Separation** – The longitudinal separation between the aircrafts operating along the same, converging, or reciprocal tracks is expressed in time or distance.
- **Lateral Separation** – The lateral separation between the aircrafts on different tracks or different geographical locations.

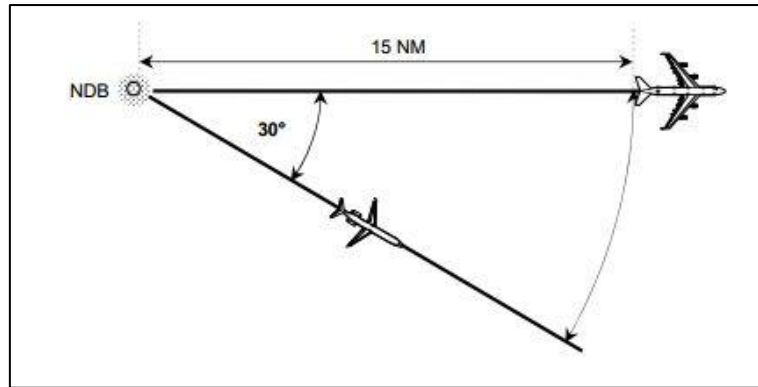
Lateral Separation Using Navigation Aids

Using the same navigation aid

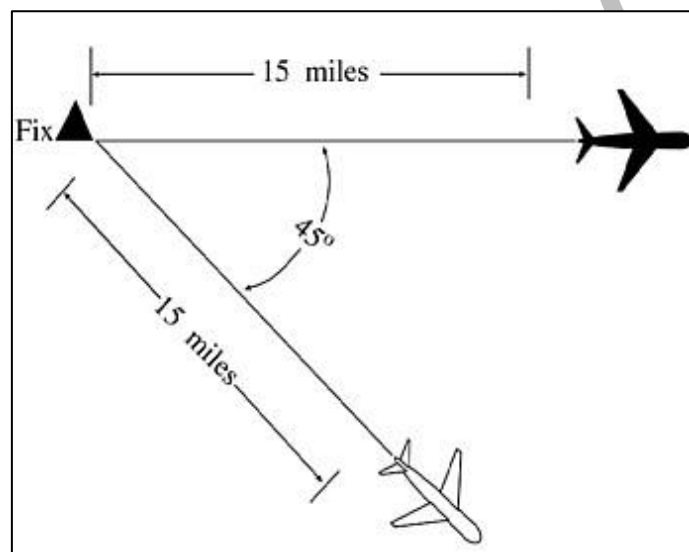
1. **VOR** (Very High Frequency Omni-directional Range): Diverging by at least **15 degrees** and at least one aircraft is at a distance of **28 km (15 NM)** or more from the facility.



2. **NDB** (Non-directional beacon): Diverging by at least **30 degrees** and at a distance of **28 Km (15 NM)** or more from the facility.



3. **DR (Dead Reckoning):** Diverging by at least **45 degrees** and at a distance of **28 Km (15 NM)**.



Longitudinal Separation

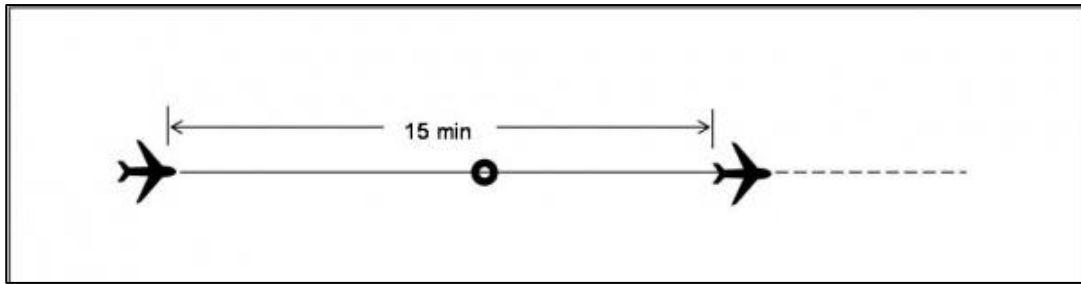
1. **Same Track** - The angular difference of which is less than **45 degrees** or more than **315 Degree**.
2. **Reciprocal Track** - The angular difference of which is more than **135 degrees** but less than **225 degrees**.
3. **Crossing Track** - Intersecting tracks or portions thereof other than those specified in a) and b) above.

Longitudinal Separation based on Time (Navigation Aids)

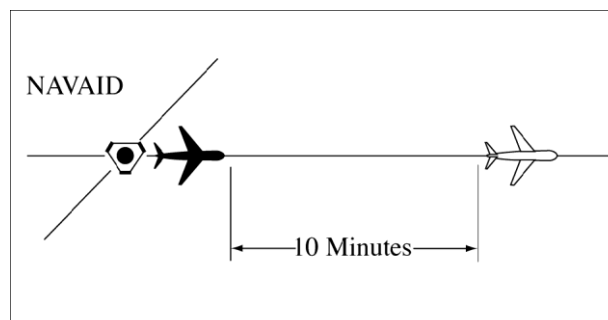
Aircraft flying on the same track:

(a) 15 minutes

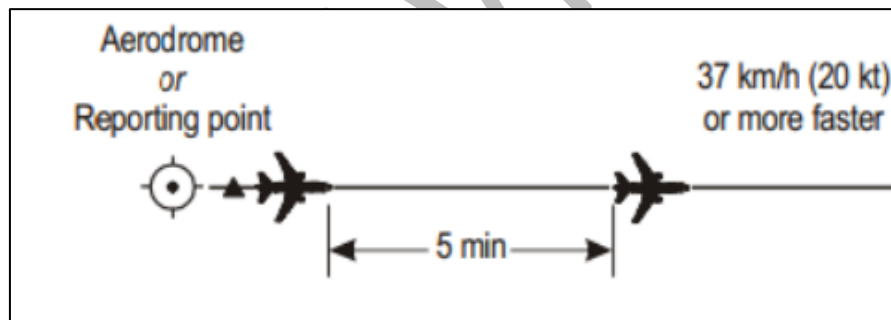
(Longitudinal separation for aircraft on the same track achieved by requiring them to pass over a specified point with separation of at least 15 minutes)



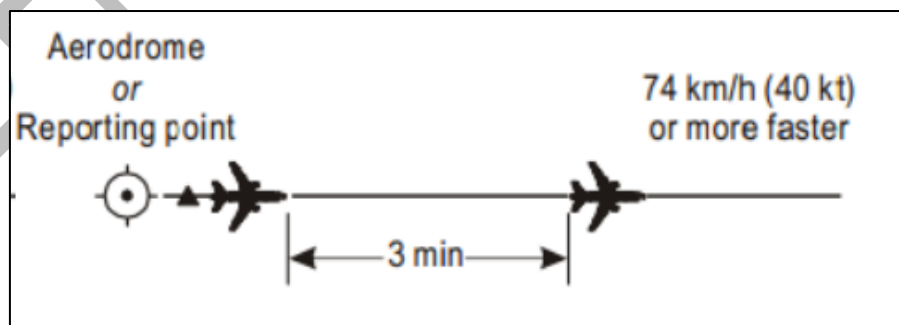
(b) **10 minutes**, if **navigation aids** permit frequent determination of position and speed



(c) **5 minutes** if the **preceding aircraft** is maintaining a true airspeed of **37 km/h (20 knots)** faster or more than the succeeding aircraft.



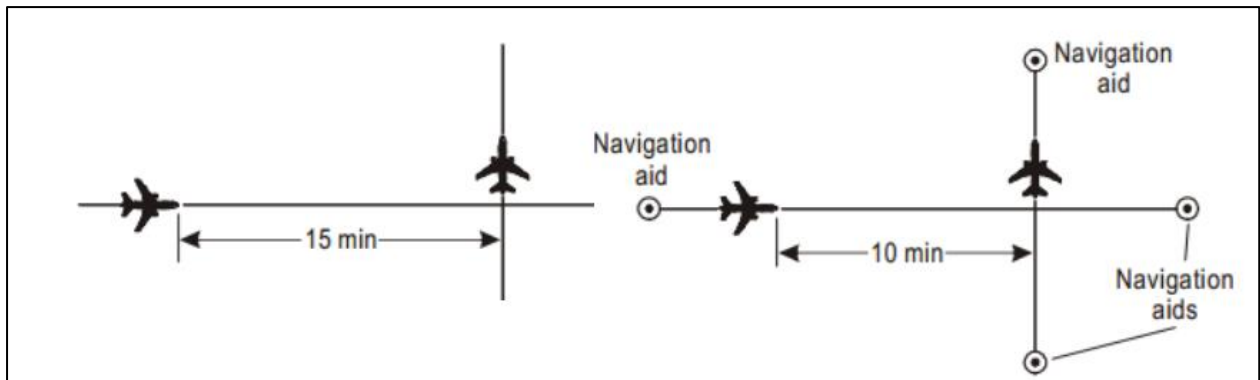
(d) **3 minutes** if the **preceding aircraft** is maintaining a true airspeed of **74 Km/h (40 knots)** faster or more than the succeeding aircraft.



Aircraft on Crossing Tracks:

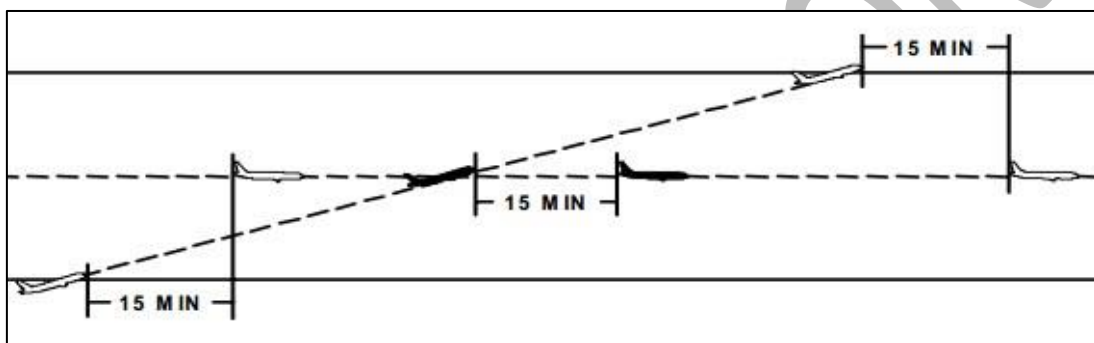
(a) **15 minutes** at the point of intersection of the tracks

(b) **10 minutes** if **navigation aids** permit frequent determination of position and speed.

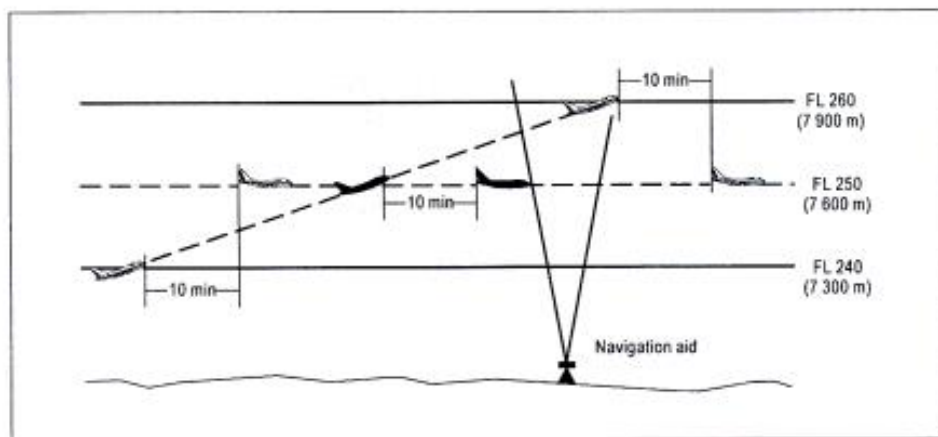


Aircraft Climbing or Descending Same Tracks:

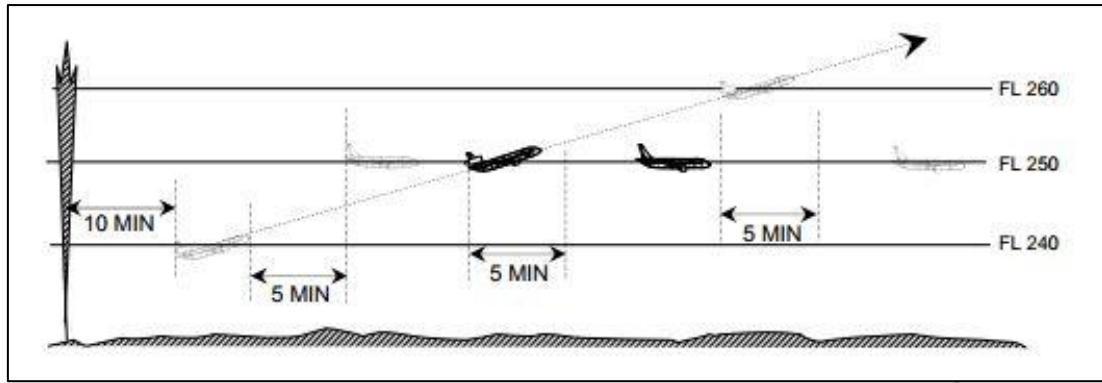
(a) **15 minutes** while vertical separation does not exist.



(b) **10 minutes** while vertical separation does not exist and only where **navigation aids** permit frequent determination of position and speed

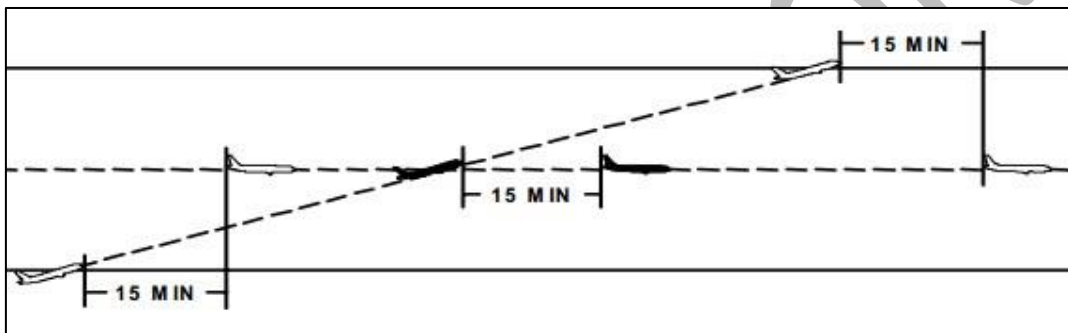


(c) **5 minutes** while vertical separation does not exist, provided that the **level change is commenced within 10 minutes of the time the second aircraft has reported over an exact reporting point.**

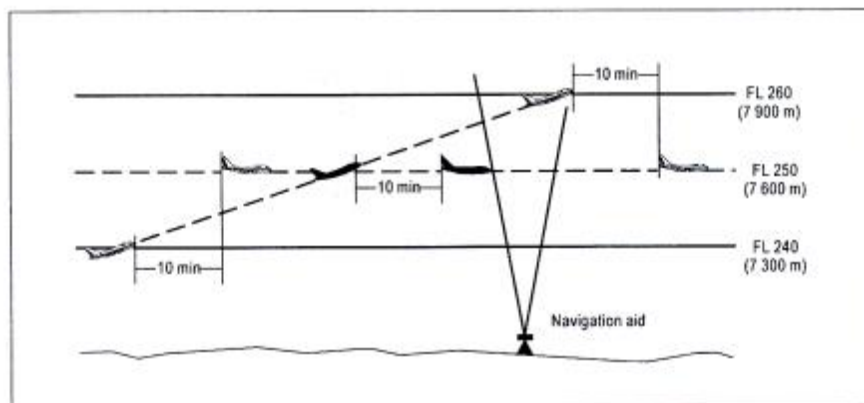


Climbing and Descending on Crossing Tracks:

- (a) **15 minutes** while vertical separation does not exist.

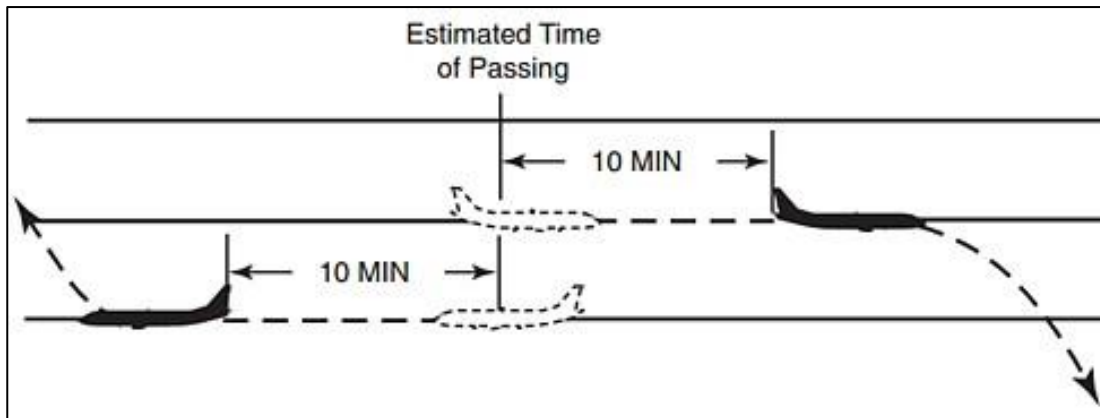


- (b) **10 minutes** while vertical separation does not exist if **navigation aids** permit frequent determination of position and speed.



Aircraft on Reciprocal Tracks

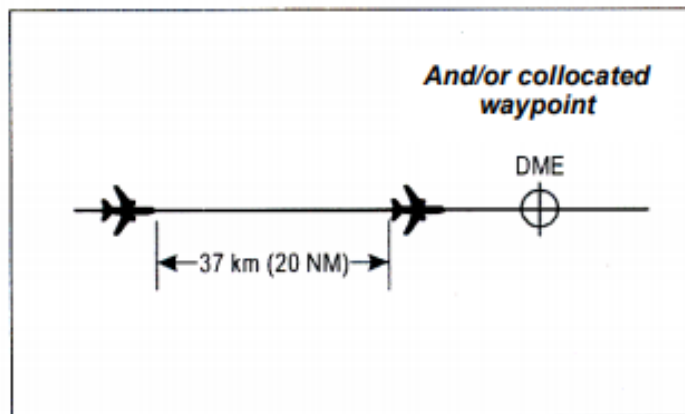
- (a) Where lateral separation is not provided, vertical separation shall be provided for **at least 10 minutes prior to and after the time the aircraft are estimated to pass, or are estimated to have passed.**



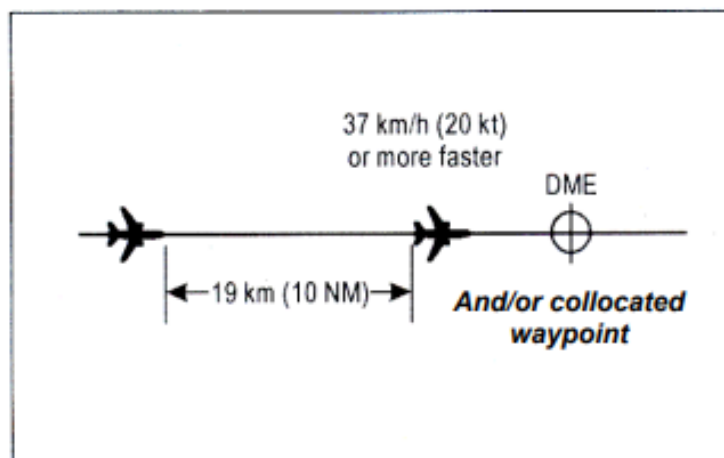
Longitudinal Separation based on Distance (DME)

Aircraft on the Same Track:

- (a) **37 km (20 NM)**, provided each aircraft utilizes "on-track" DME stations

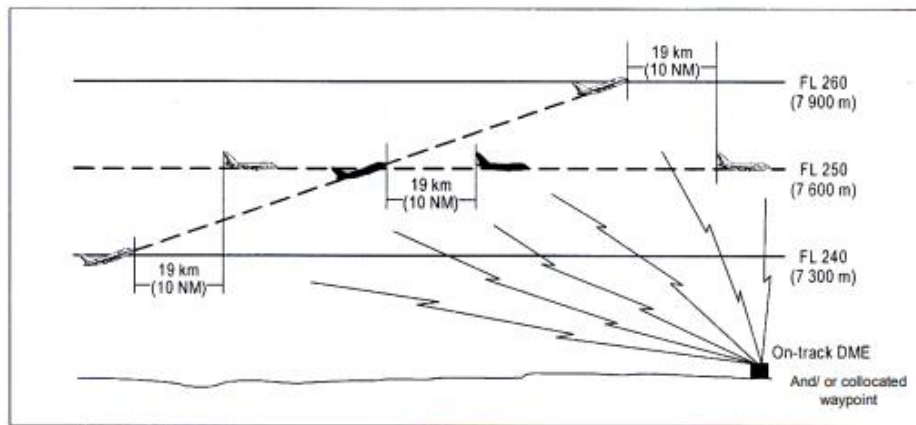


- (b) **19 km (10 NM)** provided the leading aircraft maintains a true airspeed of **37 km/h (20 knots)** faster or more than the succeeding aircraft.



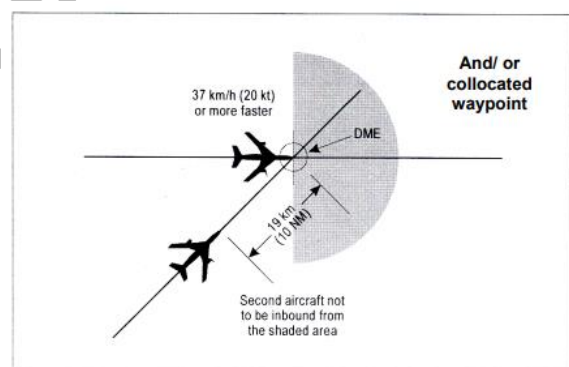
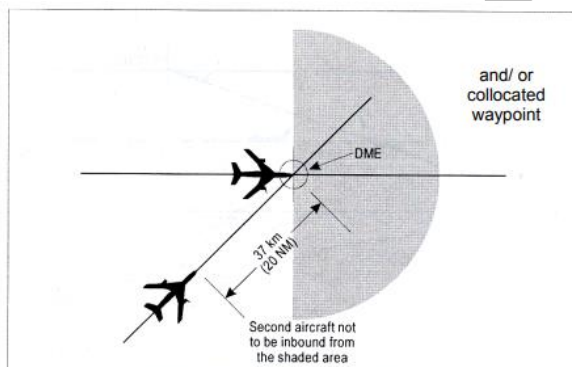
Aircraft Climbing or Descending on Same Tracks:

- (a) **19 Km (10 NM)** while vertical separation does not exist, provided each aircraft utilizes "on-track" DME stations. (Climbing aircraft comes to that level only for a few seconds)



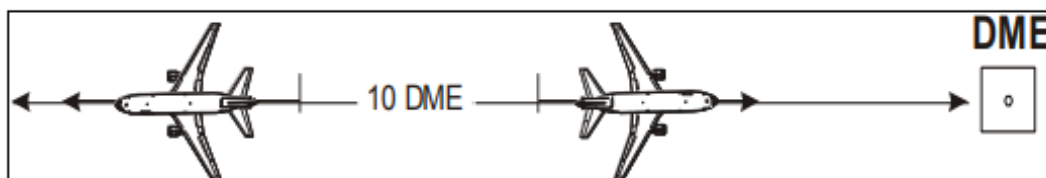
Aircraft on the **Crossing Tracks:**

- (a) **37 km (20 NM)**, provided that the relative angle between the tracks is less than 90 degrees
- (b) **19 km (10 NM)** provided the leading aircraft maintains a true airspeed of **37 km/h** (20 knots) faster or more than the succeeding aircraft.



Aircraft on **Reciprocal Track:**

- (a) At least **10 NM** apart, provided that it has been established that the aircraft have passed each other.



Longitudinal Separation Minima with Mach Number Technique based on Time and Distance

Minimum Longitudinal separation between Turbojet aircraft on same track (Level, Climbing or descending flight)

- **10 mins**
- **80 NM** – RNAV distance based separation Minima between RNAV equipped aircrafts operating on designated RNAV routes or on ATS routes defined by VOR.

Required Performances

These are the concepts which are used to determine the accuracy of different systems in flight and how well the system alerts the pilot when things are less than promised. A number attached to these concepts relates to that accuracy.

RNP (Required Navigation performance)

- RNP is a type of Performance Based Navigation (PBN) that allows the aircraft to fly a specific path between two 3-D defined points in space.
- PBN specifies that aircraft RNP and area navigation (RNAV) systems and performance requirements be defined in terms of accuracy, integrity, availability, continuity & functionality required for operations in particular airspace.
- The number attached to RNP (eg. RNP 4) is the distance in NM from intended position within which flight would be for at least 95% of total flying time.

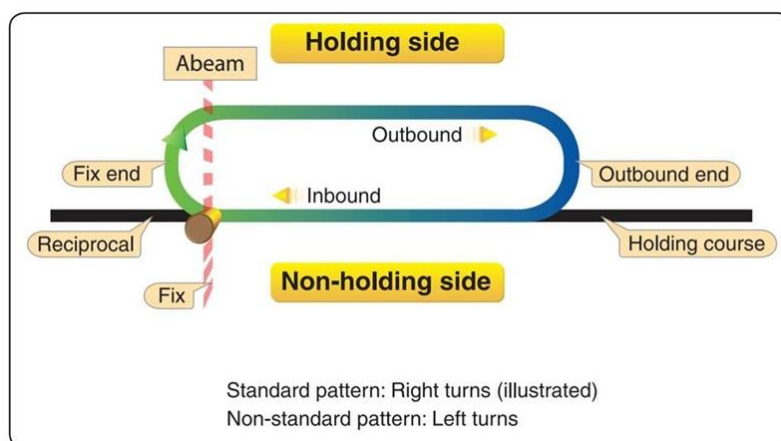
RCP (Required Communication performance)

- The number attached to RCP (eg. RCP 10) is the number of seconds it takes for an instruction to travel from ground to aircraft and acknowledgment back to the ground.

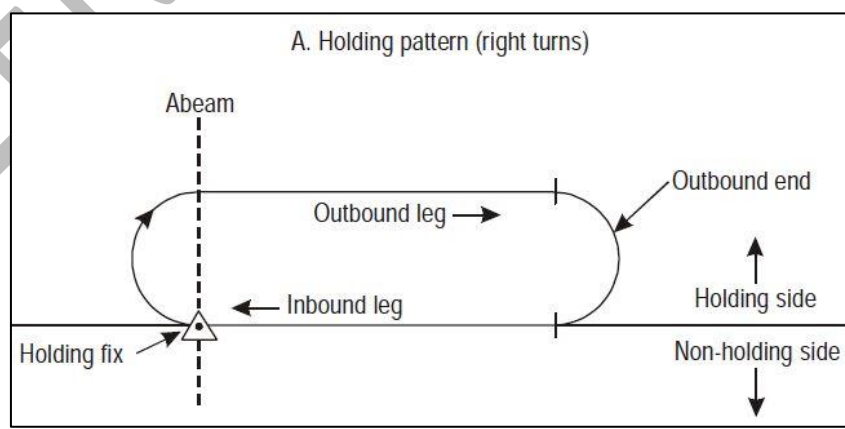
RSP (Required Surveillance performance)

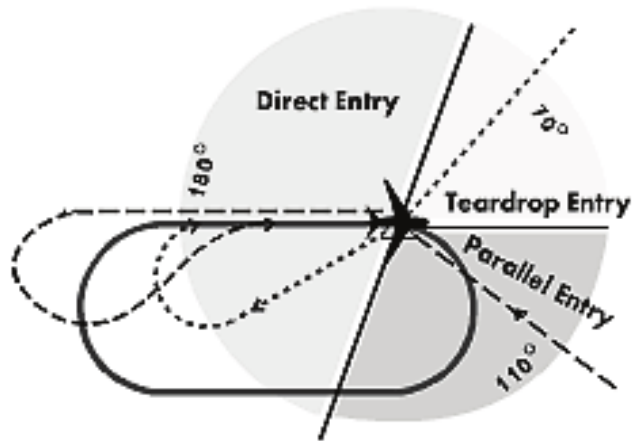
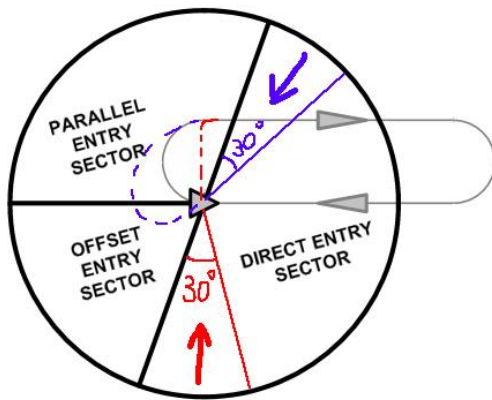
- The number attached to RSP (eg. RSP 240) is the number of seconds it takes for surveillance data from the CSP (Communication Service Provider) interface to arrive at the ATSU flight data processing system.
- RSP 240 means 99.9% of surveillance data must be delivered in less than 240 seconds.

Holding Pattern



- Entry into the holding pattern is heading based.
- All turns in the holding pattern are **3°/sec or bank angle of 25°** - whichever is lesser.
- **Standard holding patterns turns are made to right;** non-standard holding pattern turns are made to the left.
- Below 14000 ft - 1 min outbound leg.
- Zone of flexibility of the sector boundaries is 5° either side.
- Descending or climbing in a holding pattern is called shuttle.





Separation of Aircraft Holding in Flight

- **5 minutes** of flying time from holding area or **prescribed distance (5NM)**.

Wake Separation

Wake Turbulence Category of aircrafts (according to Maximum Certified Take-off Mass):

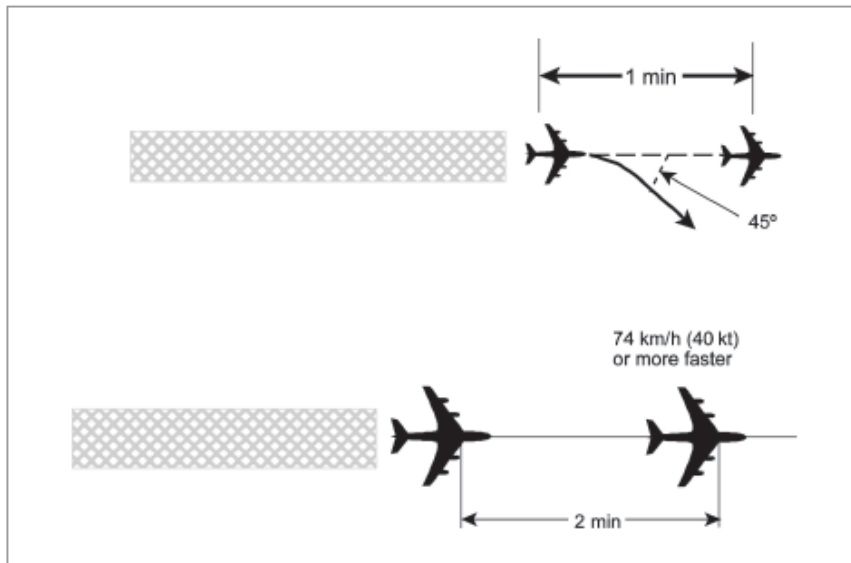
Aircraft Classification	Weight
Heavy	1,36,000 kg or more (Certified Take-off Mass) (A380- 340,000 kg)
Medium	less than 1,36,000 kg but more than 7,000 Kg (737- 78,000 kg)
Light	less than 7,000 kg

Non-Radar Wake Turbulence Longitudinal Separation Minima between aircrafts:

Separation between Departing Aircrafts -

Separation is provided **between light or medium aircraft taking-off behind a heavy aircraft**. There is no separation between heavy & heavy aircrafts.

- 1 minute separation** if aircraft are to fly on tracks diverging by at least **45 degrees immediately after take-off** so that lateral separation is provided. (Rare situation)
- 2 minutes** is required if the **tracks are less than 45 degrees apart** also if the preceding aircraft is **74 Km/h (40 Kt) or more, faster** than the following aircraft and both aircraft propose to follow the same track.



(c) The above mentioned separation rules are valid when the aircraft are taking off from the

1. Same runway
2. Parallel runway separated less than 760 m.

Departing Aircraft

Leading aircraft	Following Aircraft	Conditions	Separation Minima
Heavy	Medium or Light	a. Same runway b. Parallel runway separated by less than 760	2 Minutes
Medium	Light		
Heavy (full length take off)	Medium or Light	Departing from a. An intermediate part of the runway or b. Intermediate part of a parallel runway	3 Minutes
Medium (Full length take off)	Light		

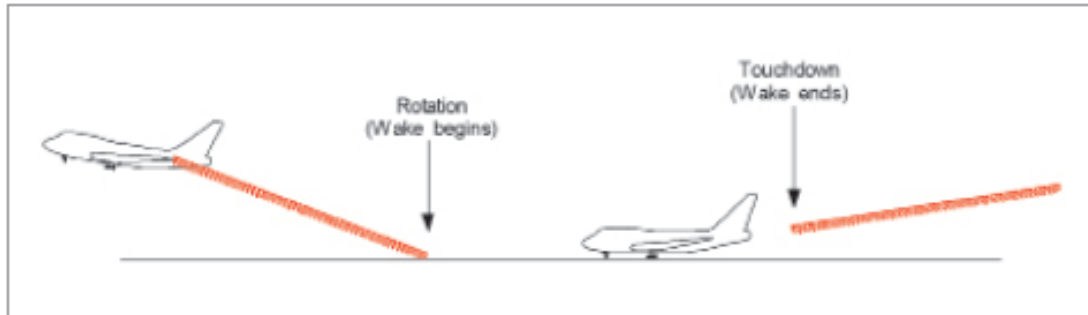
Separation between Arriving Aircraft

Leading Aircraft	Following Aircraft	Separation Required
Heavy	Medium	2 Minutes
Heavy or Medium	Light	3 Minutes

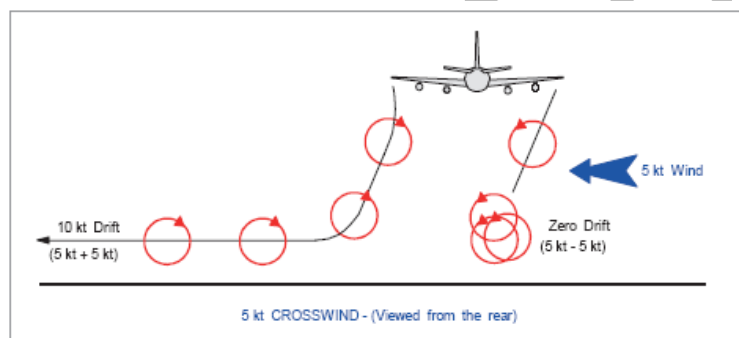
Displaced Landing Threshold

2 minutes separation shall be applied between light, medium or heavy aircraft when runway has a displaced threshold.

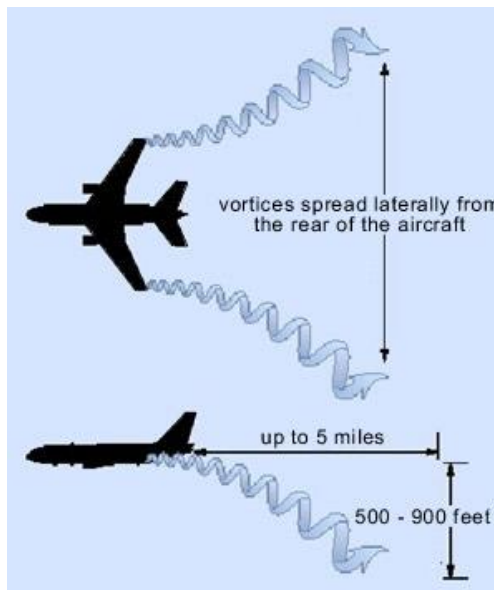
- a) When **landing behind a heavy jet you should touch down beyond his touch down point.**
- b) When **taking off behind a heavy jet you should turn/take off behind/prior to his take off point** to avoid vertical.



- c) **Light quartering tail wind** or Light cross winds (approximately 5 knots) will bring the upwind vortex on the runway for max time causing **max hazard** to incoming aircraft.



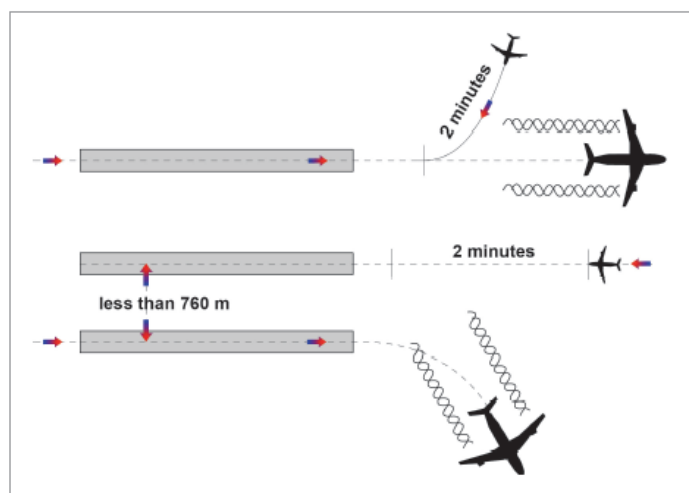
- d) **Worst vortices** are produced when aircraft is **HEAVY, CLEAN and SLOW (H.C.S.) configuration.**
- e) **Vortices circulate as outward, upward and inward clockwise on the left wing and Anti-clockwise on the right wing** when viewed from behind.
- f) **Vortices sink down** below the flight path of the aircraft that generates them.
- g) Vortices forms when aircraft generates lift.
- h) If a small aircraft experiences wake turbulence of a large aircraft, it can **induce rolling in small aircraft.**



Opposite Direction

2 minutes between a heavy and light or medium aircraft and between a medium and light aircraft when the heavier aircraft is making a low missed approach and the lighter aircraft is:

- (a) Utilizing an opposite direction runway for take-off, or
- (b) Landing on the same runway in the opposite direction, on a parallel direction runway separated by less than 760 m



Radar Separation Minima (Separation in Radar controlled environment)

Leading Aircraft	Following Aircraft	Separation
Heavy	Heavy	4 NM
Heavy	Medium	5 NM

Heavy	Light	6 NM
Medium	Light	5 NM

Radar Separation

Primary Radar	Secondary Radar
Only display aircraft symbols with no call sign	Display aircraft complete call sign along with pr. ALT or FL, speed & mode emergency
To positively identify an aircraft to manoeuvre by more than 30°	For +ve Identification pilot should press IDENT only when requested by ATC If IDENT is not working - switch off your Transponder to 'Stand by' & 'on' Ex. of Secondary Radar is Mode C Transponder - total of 4096 Codes Mode C Transponder has an accuracy of ± 300 ft

ATC has two Radar Approaches -

- 1) Primary duty of Radar unit is to provide radar separation.
- 2) **SRA (Secondary/Surveillance Radar Approach)** - SRA Approach **terminates at 2 mile** from touch down zone.
- 3) **PAR - (Precision Approach Radar)** - **Terminates at less than 1/2 mile from runway and gives distance and level information at each 0.5 NM.**
- 4) Horizontal radar separation shall be 5 NM, but on approach 6 NM.
- 5) Minimum Radar Separation ever possible is 3 NM.
- 6) ATC may ask to change speed by ± 20 kts.
- 7) On radar approach ATC (Radar Controller) should never ask aircraft to change his speed and shouldn't be interrupted for more than 5 secs in last 4 NM of approach.
- 8) Radar contact means the aircraft has been identified on the radar display and radar flight instructions will be provided until termination.
- 9) **"Resume own Navigation"** means use your Navigation facility for Navigation.
- 10) Radar controller shall issue missed approach instructions if the aircraft is not visible on the radar display for the last 2 NM of approach or if landing clearance is not issued by ATC when 2 NM from touch down, then go missed approach.
- 11) Delhi radar controller must inform non-radar controller (Jaipur) 8 NM before.
- 12) Long final means 8 NM from the approach end.
- 13) Controlled Aerodrome Means - it must have a control to the ATC service.
- 14) Fly heading 30 degrees means - 030 Magnetic Heading.
- 15) Departing aircraft may depart in any direction if the arriving aircraft is making a straight in approach and it is 5 min estimated to be over the instrument runway.
- 16) Controller should inform aircraft of mean wind speed change: 10 kts head wing change, 5 kts x-wind change, 2 kts tail wind change.

- 17) Radar controller must vector the aircraft not closer than 2.5 NM from other airspace.
- 18) Composite separation is a method in which ATC may reduce vertical & horizontal separation to Half.
- 19) An aircraft must be on glide path between 3 NM to 10 NM from runway and between 300m (984 ft) to 900m (2800 ft) on approach path.
- 20) In PSR, pilot should expect his aircraft identification on radar during departure/arrival when he is 1 NM away from end of runway.
- 21) Separation minima can be reduced when PIC has preceding aircraft in sight and is able to maintain own separation.
- 22) ICAO Doc 4444 states that a time check has to be done before commencement of taxiing for take-off and ATZ should pass the correct time to aircraft.
- 23) If SSR is found inoperative during pre-flight check and immediate repair is not possible then fly to the closest airport and repair SSR.

Mode C Transponder

- SSR - Secondary Surveillance Radar Mode C Transponder
- **Hijack or Unlawful Interference - 7500**
- **Communication failure - 7600**
- **Distress/ emergency - 7700**
- No code allotted by ATC - 2000 by default
- ATC may ask you to switch off Transponder when error exceeds +/- 300 ft
- SSR transponder to be switched on (operated all times) unless advised by ATC
- Pilots always have to read back the mode and the code to be set
- Mode A transmits call sign only
- Mode C transmits call sign and Pressure altitude
- Press IDENT key only when asked by ATC
- If IDENT key fails put it to stand by (SBY) and "ON" several times.
- No 8 and 9 keys are not used in transponder panel, 5378 is not a valid code

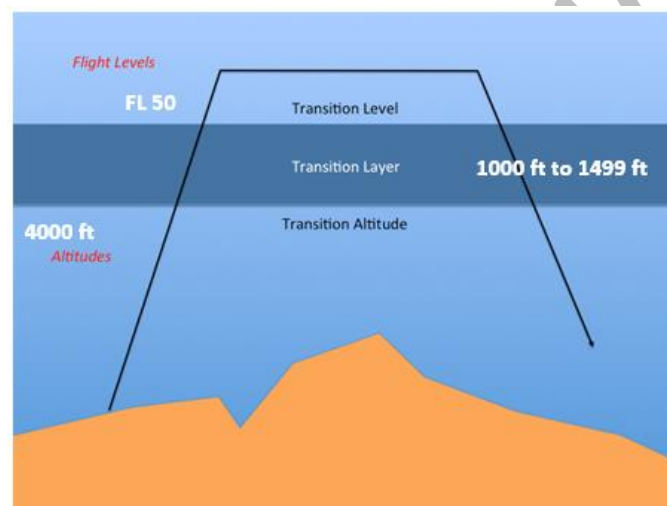
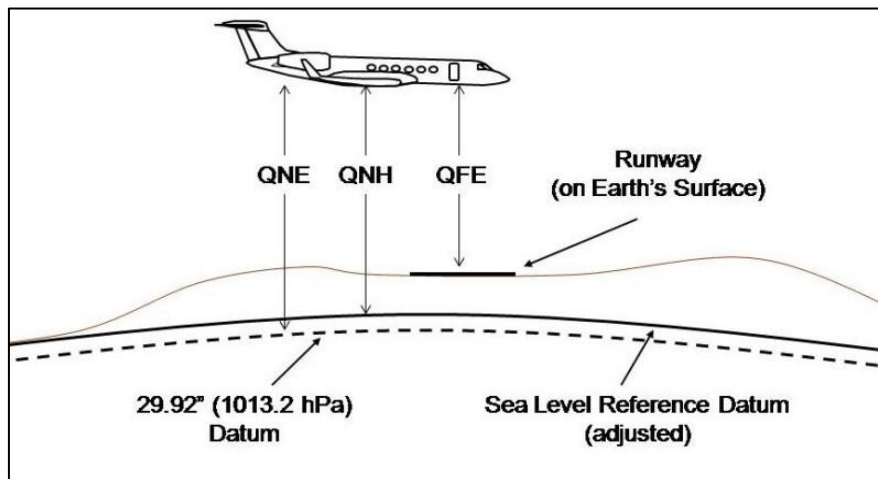
Terminology

- **ETA - Estimated Time of Arrival** (Time overhead destination)
- **EAT - Expected Approach Time** when aircraft after delays will leave the hold for Commencement of Approach
- **DER - Departure End of Runway** (Clearance is '0' ft)
- **OCA - Obstacle Clearance Altitude** (Reference from sea level)
- **MEHT - Minimum Eye Height** (Threshold Crossing Height) (50 ft)
- **OIS - Obstacle Identification Surface**

- **MSA - Minimum Sector Altitude** (25 NM)
- **TCAS - Traffic Collision Avoidance System** (CAS)
- **ACAS - Airborne CAS** (It promotes situational awareness & keeps searching for Conflicting Traffic.
- **DA (DH) - Decision Altitude** (Height) - Altitude at MAP (Missed Approach Point) in precision Approach
- **MDA (MDH) - Minimum Descend Altitude** (Height) - Altitude at MAP in Non-precision Approach

ZENITH AVIATION

Standard Altimeter Setting Procedure



- Below Transition Altitude - Vertical position will be expressed as Altitude from QNH
- Above Transition Level - Vertical position will be expressed as flight level from QNE
- Transition Layer is either 1000' or 1499'
- Delhi Airport Elevation - 770 ft from Sea Level
- While climbing through Transition Layer - Vertical Position is expressed as Flight Level (FL)
- While descending through Transition Layer - Vertical Position is expressed as Altitude
- Transition Altitude in India is 4000 ft
- Lowest Useable FL in India is FL 50 and highest is FL 460
- QNH will be communicated to Pilot in the taxi Clearance and while approaching (before Transition Level) for landing before Aircraft enters the circuit pattern.
- If QNH is not reported by MET department than QNH of nearest airport is used (min. QNH among all neighboring airport) and is called regional QNH.
- Transition Altitude is published in AIP and appropriate charts.
- As transition layer varies in summers and winters, transition layer is not published in AIP, it is confirmed with ATIS or ATC before flight.

- A pre-flight altimeter check should be carried out at a known elevation on aerodrome.
- Highest transition altitude is used as common TA if two or more aerodromes are nearby.
- Tolerance of the altimeter working with different altitudes:
 - 0 – 30000 ft : +/-60 ft
 - 0 – 50000 ft : +/-80 ft

Circle to Land

- **Circle to Land** - A visual flight manoeuvre keeping runway in sight for landing on runway not suitable for straight in Approach
- You can descend below MDA only when you are in visual contact with the ground and runway Light are in sight
- If the visual reference is lost -
 - Initiate the climb towards the landing runway
 - Initiate a published missed Approach
- Minima of Instrument Approach are decided by the state of the Aerodrome (eg. Uttarakhand), AAI, DGCA or State of operator
- State of operator is responsible for approving methods of calculating minimum flight altitudes.

Procedure Turn

Procedure Turn

A turn executed in the opposite direction of runway to permit the aircraft to proceed along a reciprocal track.

Base Turn

At the end of the outbound track and the beginning of the Intermediate fix, the turn executed towards the runway is a base turn.

NTZ (No Transgression Zone)

It is a corridor of airspace at least 610 metres wide located centrally between the two extended runway centrelines, where a penetration by an aircraft requires a controller intervention to manoeuvre any threatened aircraft on an adjacent approach. It extends from the nearer runway threshold to the point where 1000 ft vertical separation is reduced between aircrafts.

FLIGHT INFORMATION SERVICE (FIS)

A Flight information service (FIS) is to be provided to all aircraft which are likely to be affected by the information and which are:

- ☐ provided with air traffic control service; or
- ☐ otherwise known to the relevant air traffic service units;

Note: FIS does not relieve the PIC of an aircraft of any responsibilities and he/she has to make the final decision regarding any suggested alternation of flight plan.

ATSU - Air Traffic Service Unit

Precedence - When ATSU's provide both FIS and ATC service, the provision of ATC service shall have precedence over the provision of FIS whenever the provision of ATC service so requires. In certain circumstances aircraft on final approach, landing, take-off or climb may require essential information without delay, before ATC service information.

Scope of Flight Information Service - FIS includes the provision of pertinent information likely to affect safety and specifically concerning:

- ☐ **SIGMET and AIRMET**
- ☐ **pre-eruption volcanic activity, volcanic eruptions and volcanic ash clouds;**
- ☐ the release into the atmosphere of **radioactive materials or toxic chemicals;**
- ☐ changes in the serviceability of **navigation aids;**
- ☐ changes in condition of aerodromes and associated facilities, including
- ☐ information on the state of the aerodrome movement areas when they are affected by **snow, ice or significant depth of water;**
- ☐ **un-manned free balloons**

Voice - Automatic Terminal Information Service (Voice-ATIS) broadcasts are provided at aerodromes where there is a requirement to reduce to the communications load on the **ATS VHF air-ground communication channels**. When provided, Voice-ATIS broadcasts are to comprise:

- a. one broadcast serving arriving aircraft; or
- b. one broadcast serving departing aircraft; or
- c. one broadcast serving both arriving and departing aircraft

ATIS

- ATIS (Automatic Terminal Information Service) - is a voice broadcast of current weather of the aerodrome **within 5 NM (Radius) and 5000 ft (1500 m) height**
- ATIS broadcast can be on a **discreet VHF frequency or on a voice channel of VOR, but can never be on the voice channel of ILS.**

- ATIS broadcast is updated upon receipt of any official weather (regardless whether there is a content change or no) from MET department or immediately after a significant weather change.
- Preparation and dissimulation of ATIS is a **responsibility of Air Traffic Services**.
- It is a voice broadcast of essential information in a high activity terminal area.

Alerting Service

Provision of Service - It is a requirement for all contracting states of ICAO to provide SAR (Search and Rescue) facilities for all aircraft flying in the airspace of that state. In order to alert the SAR (and other services- fire, ambulance, police, mountain rescue, civil defense), states are required to have a formal system.

How does ATS know that there is an emergency?

Uncertainty Phase (INCERFA)

Except when no doubt exists as to the safety of the aircraft and its occupants, the uncertainty phase is declared when:

- ☐ **No communication** has been received from an aircraft within a **period of thirty minutes** after the time a communication should have been received, or thirty minutes from the time an unsuccessful attempt to establish communication with the aircraft was first made, whichever is the earlier, or when
- ☐ **An aircraft fails to arrive within thirty minutes** of the estimated time of arrival last notified to, or estimated by ATCUs, whichever is the later

Alert phase (ALERFA)

When **apprehension exists as to the safety of the aircraft and its occupants**, or when the distress phase is more appropriate. The alert phase is declared when:

- ☐ Following the uncertainty phase, subsequent attempts to establish communication with the aircraft have failed to reveal any news of the aircraft;
- ☐ **An aircraft has been cleared to land and fails to land within five minutes** of the estimated time of landing and communication has not been re-established with the aircraft, or when
- ☐ Information has been received which indicates that the **operating efficiency of the aircraft has been impaired**, but not to the extent that a forced landing is likely, or
- ☐ **An aircraft is known or believed to be the subject of unlawful interference.**

Distress phase (DETRESFA)

When there is reasonable certainty that the **aircraft and its occupants are threatened by grave and imminent danger and do require immediate assistance**, the distress phase is declared when:

- ☐ Following the alert phase further unsuccessful attempts to establish communication with the aircraft point to the probability that the aircraft is in distress, or when
- ☐ The **fuel on board is considered to be exhausted**, or to be insufficient to enable the aircraft to reach safety, or when
- ☐ Information is received which indicates that the operating efficiency of the aircraft has been impaired to the extent that a **forced landing is likely** or when
- ☐ Information is received or it is reasonably certain that the **aircraft is about to make or has made a forced landing**.

AERONAUTICAL INFORMATION SERVICE

(AIS) - Annexure 15

Aeronautical Information Publication (AIP) - A publication issued by or with the authority of a State and containing aeronautical information of a lasting character essential to air navigation. Printed by AAI

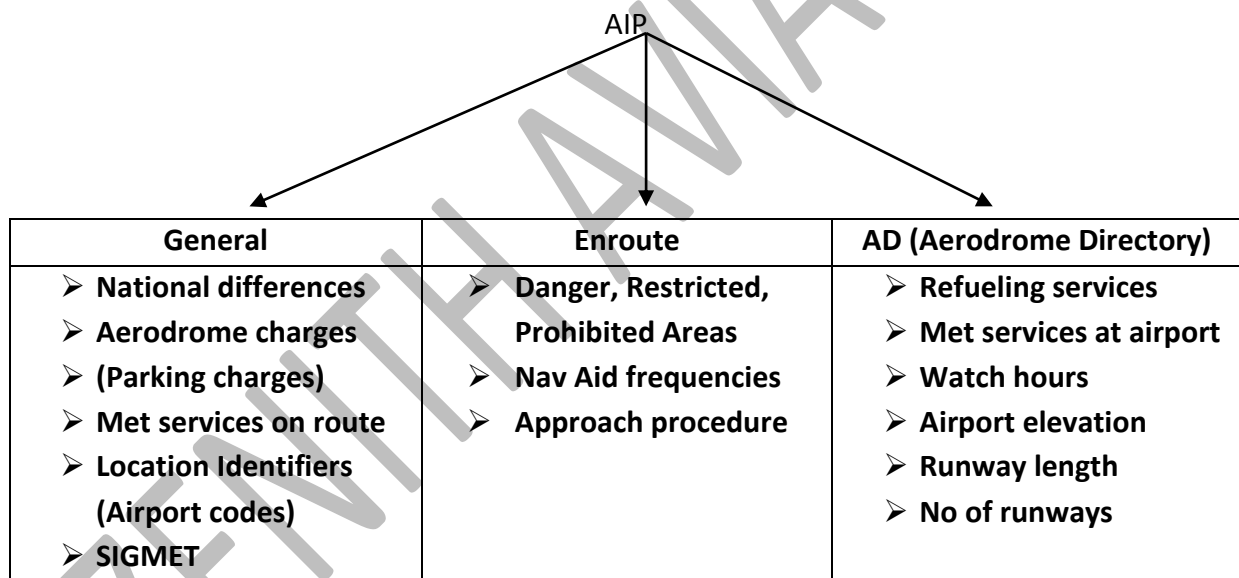
Objectives of the AIS - The objective of the aeronautical information service is to ensure the flow of information necessary for the safety, regularity and efficiency of international air navigation. (Foreign Pilots)

Contents - An AIP shall consist of the three parts:

Part 1 - General (GEN)

Part 2 - En-route (ENR)

Part 3 - Aerodrome Directory (AD)



An Aeronautical Information Publications is to include General (GEN) the following sections and information: (DGCA)

- ☐ A statement of the competent authority responsible for the air navigation facilities, services covered by the AIP
- ☐ The general conditions under which the services of facilities are available for international use;
- ☐ A list of significant differences between the national regulations and practices of the State and the related ICAO Standards, Recommended Practices and Procedures given in

a form that would enable a user to differentiate readily between the requirements of the State and the related ICAO provisions

- The choice made by a State in each significant case where an alternative course of action is provided for in ICAO standards, Recommended Practices and Procedures.

AIP Amendment: Permanent changes to the information contained in the AIP.

AIP Supplement: Temporary changes to the information contained in the AIP which are published by means of special pages.

- **Long term supplement - More than 3 month**
- **Short term supplement - Less than 3 months**

AIRAC: Acronym (**Aeronautical Information Regulation and Control**) Signifying a **system aimed at advance notification**, based on common effective dates of circumstances that necessitate significant changes in operating practices. Example: Change from NDB to a VOR at any station.

Information concerning the circumstances listed below shall be distributed under the regulated system (AIRAC), basing establishment, withdrawal or significant changes on a series of common effective dates at **intervals of 28 days**.

The information shall be distributed by the AIS unit at least **42 days in advance** of the effective date and the **information notified shall not be changed for at least another 28 days after the effective date**.

Information to be notified by AIRAC

The establishment, withdrawal of, and premeditated significant changes (including operational trials) to:

Limits (horizontal and vertical), regulations and procedures applicable to:

1. Flight information regions;
2. Control areas;
3. Control zones;

AIRAC provides information regarding:

- Position, frequencies, call signs, known irregularities and maintenance periods, radio navigation aids and communications facilities
- Holding and approach procedures, arrival and departure procedures, noise abatement procedures and any other pertinent ATC procedures.
- Meteorological facilities (including broadcasts) and procedures;

- Runways and Stopways

Notices to Airmen (NOTAM)

Definition: A NOTAM is a **notice distributed by means of telecommunications** containing **information concerning the establishment, condition or change in any aeronautical facility, service, procedure or hazard, the timely knowledge of which is essential to personnel concerned with flight operations.**

NOTAMs are originated by the international NOTAM office Mumbai, Kolkata, Delhi and Chennai. The NOTAMs listed for international distribution are promulgated in series A, B, C, D and G as follows:

Series A: Contain information regarding changes/unserviceability of aeronautical facilities, likely to last for **more than 2 hours**.

Series B: Contains information regarding changes/unserviceability likely to last for more than 30 minutes but **less than 2 hours**.

Series C: Contain information about **domestic flights**.

Series D: Contain information regarding changes, unserviceability of aeronautical facilities for **Defense - controlled aerodromes** only.

Series G: Contain information of **general lasting character** affecting aircraft operations in general. This series is operated by the International NOTAM office Delhi and issued under DGCA.

SNOWTAM - Information **concerning snow, ice and standing water** on aerodrome pavement areas is to be reported by SNOWTAM.

ASHTAM - Information concerning an operationally significant **change in volcanic activity, a volcanic eruption and/or volcanic ash cloud** are reported by means of an ASHTAM.

Aeronautical Information Circular (AIC): A notice containing information that does not qualify for the origination of a NOTAM or for inclusion in AIP, but relates to flight safety, air navigation, technical etc...

Example: Advice on medical matters of special interest to pilots and exam dates etc.

Each contracting state shall provide AIS which includes - AIP, AIP supplements, AIP Amendment, NOTAM, Pre-flight Information Bulletin (PIB), AIC, checklist and summary. All this together is known as Integrated Aeronautical Package.

Checklist is printed every month.

ANNEXURE - 14- AERODROMES

Aerodrome - Any area of land (area or space, whether on the ground, on the roof of a building or elsewhere) or water designed, equipped, set apart or commonly used for affording facilities for the landing and departure of aircraft capable of descending or climbing vertically.

Aerodrome Elevation - The elevation of the highest point of the landing area

Aerodrome Reference Point - The aerodrome reference point is the geographical location of the aerodrome and the reference centre of its traffic zone where an ATZ is established.

Apron - A defined area on a land aerodrome provided for the stationing of aircraft for the embarkation and disembarkation of passengers, the loading and unloading of cargo, and for parking. (eg. Gate 22 C)

Maneuvering Area - That part of an aerodrome provided for the take-off and landing of aircraft and for the movement of aircraft on the surface, excluding the apron and any part of the aerodrome provided for the maintenance of aircraft.

Movement Area - That part of an aerodrome intended for the surface movement of aircraft including the maneuvering area, aprons and any part of the aerodrome provided for the maintenance of aircraft.

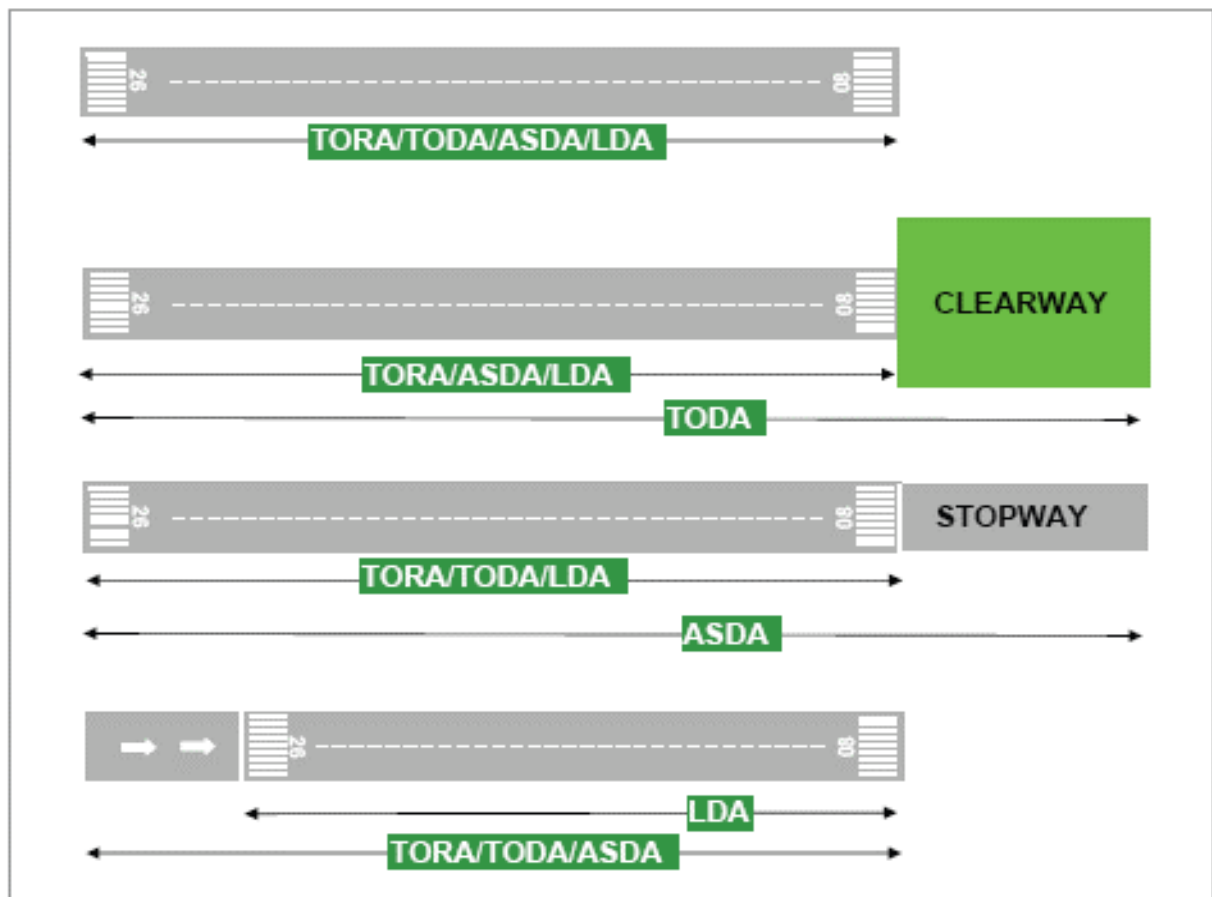
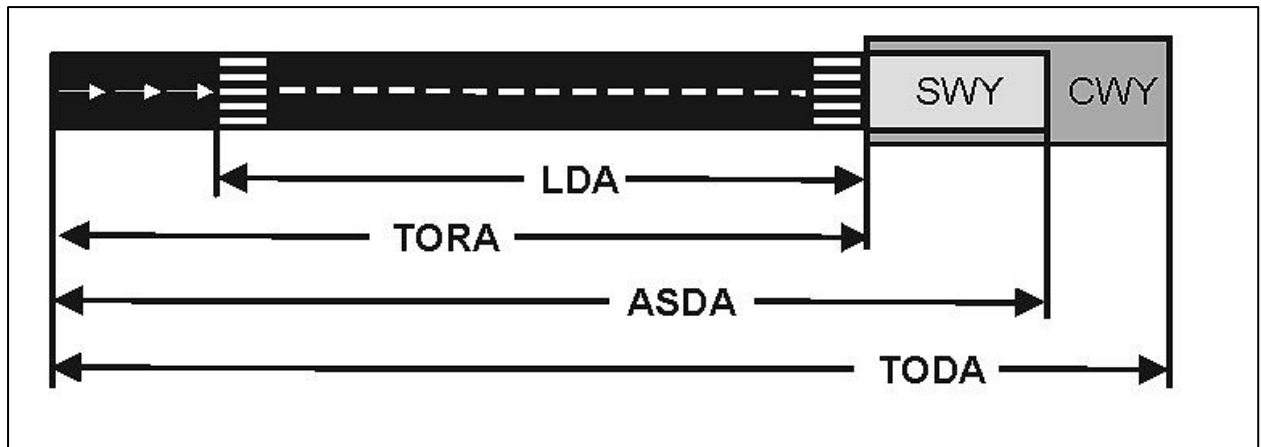
Aerodrome Traffic - Aerodrome Traffic is the aircrafts on the maneuvering area & the aircraft flying in the vicinity.

Declared Distances - The following distances shall be calculated to the nearest meter or foot for a runway intended for the use by international commercial air transport:

- ☐ **Take-off run available; (TORA)**
- ☐ **Take-off distance available; (TODA)**
- ☐ **Accelerate-stop distance available; and (ASDA)**
- ☐ **Landing distance available. (LDA)**

Clearway - An area at the end of the take-off run available and under the control of the aerodrome licensee, selected or prepared as a suitable area over which an aircraft may make a portion of its initial climb to a specified height (35ft.)

Stopway - A defined rectangular area at the end of the take-off run available prepared and designated as suitable area in which an aircraft can be stopped in the case of a discontinued take-off.



Stopway

1. Width of stopway is same as runway.
2. All four sides of stopway have steady red Lights.
3. Stopway can't be used for normal ops, going on a stopway is considered as incident and reported to DGCA within 48 hours.
4. Stopway is in centreline with runway and has yellow chevrons.
5. $ASDA = TORA + \text{Stopway}$

Clearway

1. Width 500ft, slope not exceeding 1.25%, length not exceeding 50% of TORA.
2. Area clear of obstacles upto a height of 35ft in which aircraft makes initial portion of climb
3. TODA = TORA + clearway

Aerodrome Reference Code

The reference code is used for aerodrome planning purposes, is a simple method of interrelating the specifications concerning the characteristics of aerodromes, so as to provide aerodrome facilities that are suitable for the aeroplanes that are intended to operate at the aerodrome.

Elements - The code is composed of two elements which are related to aeroplane performance characteristics and dimensions.

Element 1 is a number based on the aeroplane reference field length.

Aeroplane reference field length is defined as "the minimum field length required for take-off at maximum certificated take-off mass, at sea level, in International Standard Atmosphere conditions in still air and with zero runway slope as documented in the Aircraft Flight Manual (AFM) or equivalent document.

Element 2 is a letter based on the aeroplane wing span and outer main gear wheel span.

Code Element 1		Code Element 2		
Code number	Aeroplane reference field length	Code letter	Wing span	Outer main gear wheel span *
1	Less than 800m	A	Up to but not including 15 m	Up to but not including 4.5 m
2	800 m or more, but less than 1200 m	B	15 m or more, but less than 24 m	4.5 m up to but not including 6 m
3	1200 m or more, but less than 1800 m	C	24 m or more, but less than 36 m	6 m up to but not including 9 m
4	1800 m or more	D	36 m or more, but less than 52 m	9 m up to but not including 14 m
		E	52 m or more, but less than 65 m	9 m up to but not including 14 m
		F	65 m up to but not including 80m ⁽²⁾	14 m up to but not including 16 m

*Distance between the outside edges of the main gear wheels.

Pavement Strengths - Where pavements are used by aircraft with maximum take-off mass greater than 5700 kg, the strength of the pavement is reported by the aircraft classification number - pavement classification number (ACN-PCN) system.

- ✓ Pavement type for ACN-PCN determination;
- ✓ Subgrade strength category
- ✓ Maximum allowable tire pressure value
- ✓ Technical Evaluation method

The pavement classification number (PCN) reported shall indicate that an aircraft with an **aircraft classification number (ACN) equal to or less than the reported PCN can operate on the pavement** subject to any limitation on the tire pressure, or aircraft all-up mass for specified aircraft type(s).

Pavement type for ACN-PCN determination:	Code
Rigid Pavement	R
Flexible (or Composite) Pavement	F

Subgrade strength category:	Code
High strength : characterized by $K = 15 \text{ MN/m}^3$	A
Medium strength : characterized by $K = 80 \text{ MN/m}^3$	B
Low strength : characterized by $K = 40 \text{ MN/m}^3$	C
Ultra low strength : characterized by $K = 20 \text{ MN/m}^3$	D

Maximum allowable tire pressure category:	Code
High : No pressure limit	W
Medium : pressure limited to 1.50 MPa (Mega Pascals)	X
Low : pressure limited to 1.00 MPa	Y
Very low : pressure limited to 0.50 MPa	Z

Technical Evaluation Method	Code
Representing a specific study of the pavement characteristics and application of pavement behavior technology.	T
Using aircraft experience: representing knowledge of the specific type and mass of aircraft satisfactorily being support under regular use.	U

1. PCN 80 / R / B / W / T

If the bearing strength of a rigid pavement, resting on a medium strength subgrade, has been assessed by technical evaluation to be PCN 80 and there is no tire pressure limitation.

2. PCN 50/F/A/Y/U

If the bearing strength of the composite pavement, behaving like a flexible pavement and resting on a high-strength sub-grade has been assessed by using aircraft experience to be PCN 50 and the maximum tyre pressure allowable is 1.00 MPa.

RUNWAYS

Usability

For an aerodrome to be used for commercial transport, the schedules agreed must be able to be operated. This means that the runways are not only long enough and strong enough for the aeroplanes to use, but also sited to take into account prevailing meteorological conditions, air traffic requirements, and obstacle clearance. Wind is probably the most important meteorological consideration and runway direction must be based towards the prevailing wind. It is a requirement of ICAO that an aerodrome must be useable for not less than 95% of the notified hours of operation Called as **Usability Factor**.

Types of Runway

Runways are described by the types of operations that can be conducted on the runways:

Non Instrument (Visual) Runway

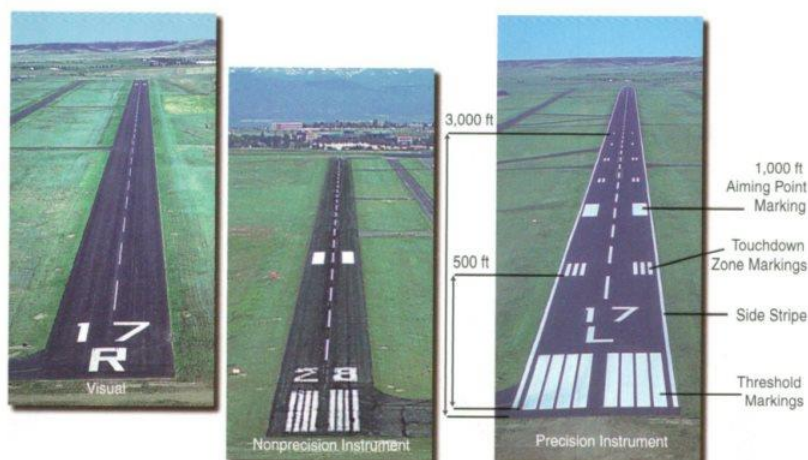
The take-off and landing criteria are determined visually with reference to ground visibility, RVR cloud ceiling, and day/night conditions.

Instrument Runway

A runway to which instrument arrival and departure procedures are applied is called an instrument runway.

There are two types of instrument runways:

- ☐ **Precision runways:- CAT I, CAT II & III**
- ☐ **Non-precision runways.**



Take-off Runway

A runway used only for take-off operations usually due to excessive obstacles precluding a useful approach.

Length of Runways - The actual length of a runway should be adequate to meet the operational requirements of the aeroplanes for which the runway is intended and **should not be less than the longest length calculated to correct for local conditions** (elevation, temperature, runway slope, humidity and surface characteristics).

Width of Runways - The width of a runway should not be less than the dimension specified in the table below.

Code Number	Code Letter					
1	A	B	C	D	E	F
2	18 m	18 m	23 m			
3	23 m	23 m	30 m			
4	30 m	30 m	30 m	45 m		
			45 m	45 m	45 m	60 m

For a precision approach runway, the width should not be less than 30 m when the Aerodrome Code Number is 1 or 2.

RUNWAY MARKINGS

Runway Designator Markings - A runway designation marking shall consist of a two digit number. On parallel runways it shall be supplemented with a letter.

- ☐ For two parallel runways "L" "R".
- ☐ For three parallel runways "L" "C" "R".
- ☐ For four parallel runways "L" "R" "L" "R" (in this case the QDM for one pair will be increased by 1 to differentiate that pair from the other).

Runway Centre Line Marking - A runway centre line marking shall be provided on a paved runway. The centre line marking shall be located along the centre line of the runway between the runway designation markings.

Characteristics-

- A runway centre line marking shall consist of a line of uniformly spaced stripes and gaps.
- The length of a stripe plus a gap shall not be less than 50m or more than 75m
- The length of each stripe shall be at least equal to the length of the gap or 30m, whichever is greater.

Touch down Zone Markings - Touchdown zone markings consist of pairs of rectangular markings symmetrically placed about the runway centre line with the number of pairs related to the landing distance available. These are 150m spaced longitudinally and start from 150m from runway threshold.

Landing distance available or the distance between thresholds	Pair(s) of Markings
less than 900m	1
900m up to but not including 1200m	2
1200m up to but not including 1500m	3
1500 up to but not including 2400m	4
2400m or more	6

The aiming point marking shall commence no closer to the threshold than the distance indicated in the appropriate, except that on a runway equipped with a visual approach slope indicator system, the beginning of the marking shall be coincident with the visual approach slope origin.

Location of Aiming Point Marking

Location and dimension	Landing distance available			
	Less than 800m	800m up to but not including 1,200 m	1,200 m up to but not including 2,400m	2400m and above
Distance from threshold to beginning of marking	150 m	250 m	300 m	400 m

Threshold Marking - A threshold marking is required to be provided at the thresholds of paved instrument runways, and of paved non-instrument runways where the code number is 3 or 4 and the runway is intended for use by international commercial air transport (CAT)

Characteristics - A runway threshold marking is a pattern of longitudinal stripes of uniform dimensions disposed symmetrically about the centre line of the runway.

Threshold Strips	Runway Width
4	18 m
6	23 m
8	30 m
12	45 m

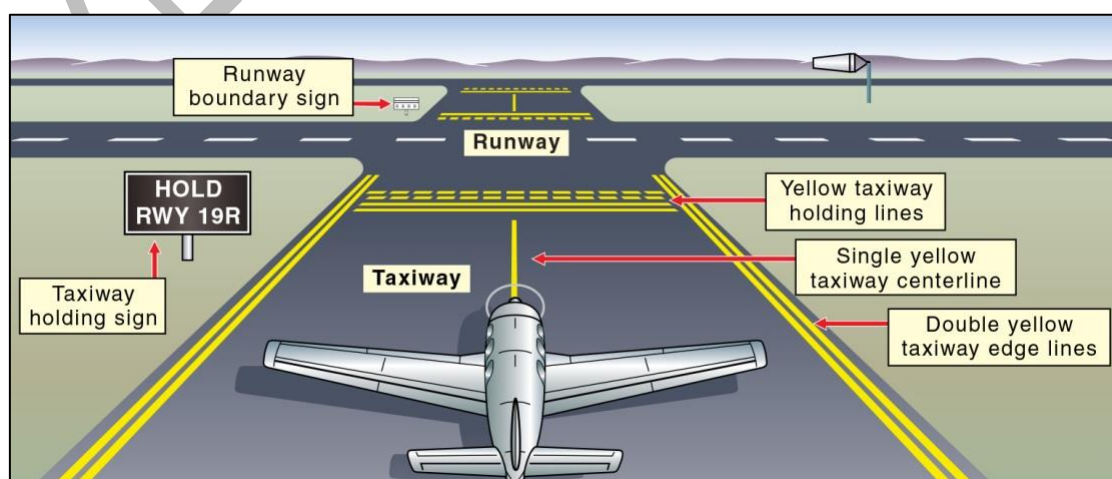
16 (8 pairs)	60 m
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Taxiway Markings

- ☐ Taxiway markings and aircraft stand markings are yellow.
- ☐ Taxiway centre line marking shall be provided on a paved runway when the runway is part of a standard taxi- route and:
 - i. There is no runway centre line marking; or
 - ii. Where the taxiway centre line is not coincident with the runway centre line.
- ☐ Taxiway markings are yellow continuous line
- ☐ Taxiway lines can be painted on runway also if runway is a part of a taxiway route but runway markings will overlap taxiway marking.

Runway Holding Position Marking

- A runway holding position marking is placed at runway holding position. The holding point itself is indicated by the mandatory sign(s) at the side of the taxiway.
- Two dashed lines are painted near the runway and solid line away from runway, **called Hold Short lines.**
- If hold short marking are not painted aircraft must stop 50 m away from runway.
- No part of the aircraft must cross the hold short line.



AERODROME LIGHTING

An aerodrome at night is a profusion of lights. Apart from the main stadium lighting around the apron and passenger areas, the runway are lit as the taxiways; building and vehicles have lights on them as do aircraft. All the coloured lights have a purpose and the arrangements of lights also have meaning.

Aerodrome Beacon

- ☐ Flashing white and green for civilian land airports
- ☐ Flashing white and yellow for a water airport
- ☐ Flashing white, yellow, and green, for a heliport
- ☐ Two quick, white flashes followed by a green flash identifies a military airport.

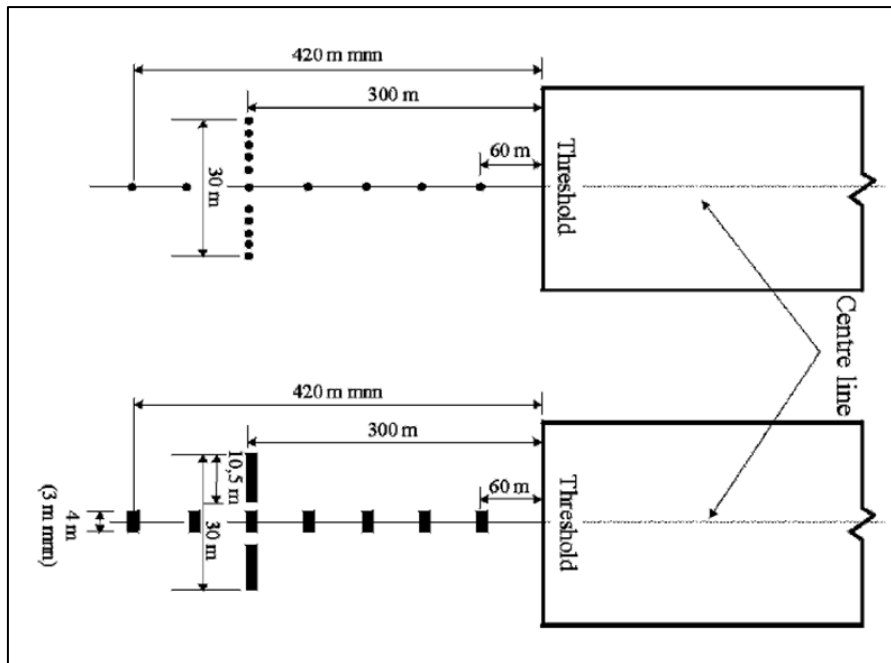
An aerodrome beacon is usually situated on top of the control tower or at an elevated position elsewhere on the aerodrome.

Approach Lighting Systems

- Simple Approach Lighting System
- Precision Approach Cat I Lighting System
- Precision Approach Category II and III Lighting System

Simple Approach Lighting System

- The system consists of a **single light source centre line extending from the threshold of the runway along the approach path for 420 m.**
- A single crossbar minimum 18 m wide and maximum 30 m wide is placed at a point 300 m from the threshold.
- The lights are fixed (not flashing) variable intensity white, showing towards an approaching aircraft.
- The centreline may be made up of barrettes not more than 3 m wide.
- The system may be used for a non-precision instrument runway.



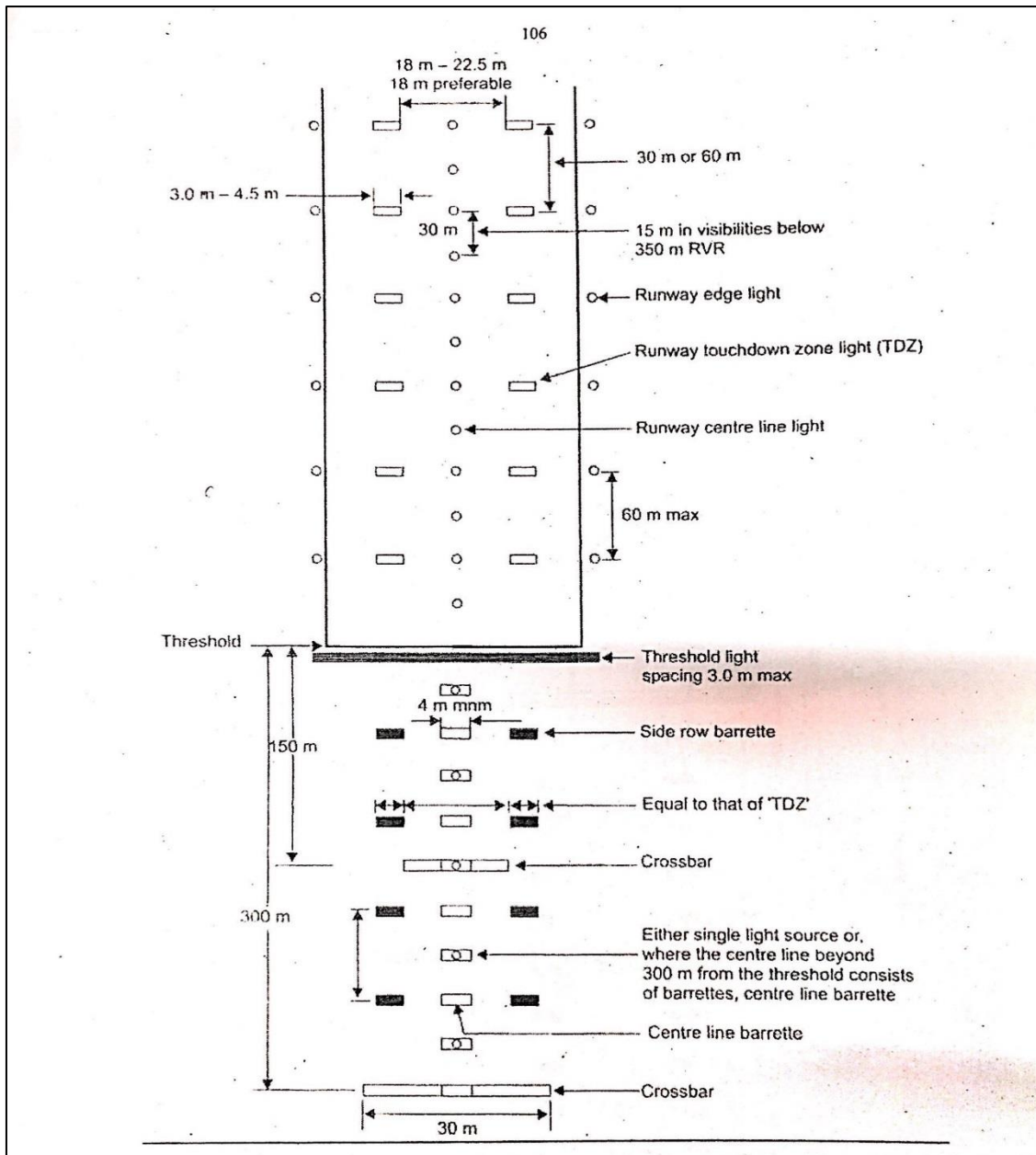
Barrettes - ICAO specifies standards for approach lighting based on the use of barrettes (or small bars). A barrette consists of a line of 4 or 5 closely spaced lights forming a small bar.

Precision Approach Category I Lighting System

- A precision approach category I lighting system shall consist of a **row of lights on the extended centre line of the runway extending, wherever possible, over a distance of 900m from the runway threshold**
- Row of lights forming a crossbar 30m in length at a distance of 300m from the runway threshold.
- Fixed lights showing variable white.

Calvert CAT II & III

- The **centreline must be exactly 900 m long**.
- The spacing between the crossbars is 150 m.
- The lights forming the centre line shall be placed at longitudinal intervals of 30m with the innermost lights located 30m from the threshold.
- The lights forming the side rows shall be placed on each side of the centre line, at a longitudinal spacing equal to that of the centre line lights and with the first light located 30m from the threshold.
- The lateral spacing between the innermost lights of the side row shall be not less than 18m nor more than 22.5m

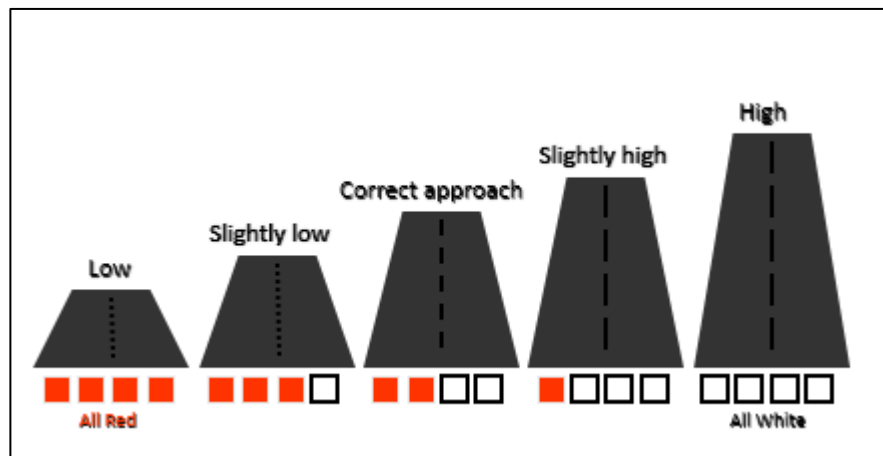


Precision Approach Path Indicator (PAPI)

PAPI are the latest generation of visual approach slope indicators which are used to give the pilot a **visual indication of the vertical deviation from the nominal glide path.**

- When on or close to the approach slope, see the **two units nearest the runway as red** and the two units farthest from the runway as white;
- When above the approach slope, see the **one unit nearest the runway as red** and the three units farthest from the runway as white; and when further above the Approach slope, see all the units as white;

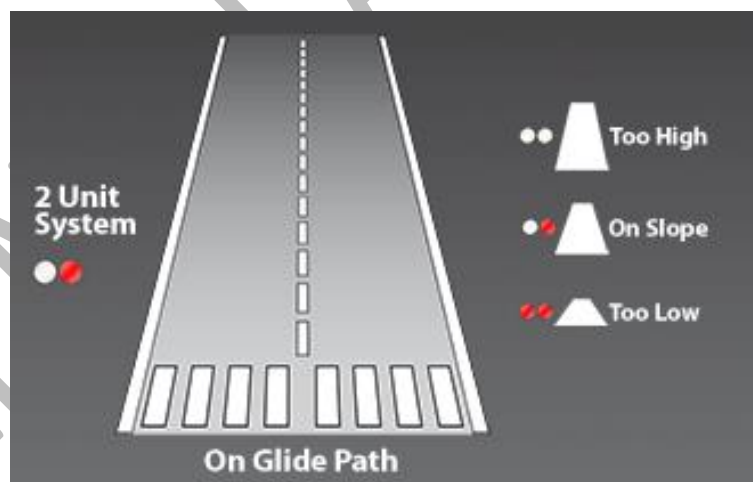
- When **below the approach slope**, see the **three units nearest the runway as red** and the **unit farthest from the runway as white**; and **when further below the approach slope**, see **all the units as red**.



APAPI

The APAPI uses only two light sources and gives indication:

- Two white - above the glide path
- One white one red - on the glide path
- Two red - below the glide path



RUNWAY LIGHTS

Runway Edge Lights

- Runway Edge lights are used on runways used at night or in low visibility operation (RVR less than 800 m).
- They are fixed (not flashing), **white**, variable in intensity and unidirectional (facing the approach direction)
- For a CAT I runway, the last 600 m or 1/3 of the runway length, the edge lights are possibly yellow, indicating the proximity of the end of the runway. This is called a caution zone.

Runway Threshold and Wing Bars

Threshold lights are provided for runways with edge lights and additional wing bars where there is a displaced threshold. They are **fixed, green, variable intensity**, and unidirectional facing up the approach.

Runway End Lights

- Runway end lights shall be **fixed unidirectional lights showing red** in the direction of the runway.
- **Where a stopway is created at the end of the runway all four sides of the stopway are outlined in red lights.**

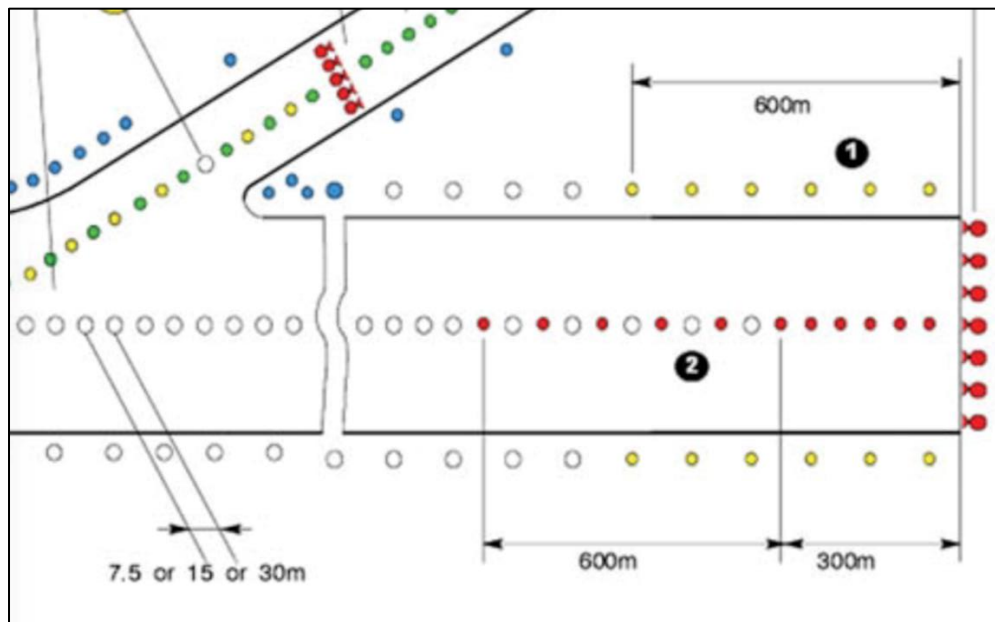
Runway Centre line Lights

- Centreline lights are **required on CAT II/III runways**.
- They are **fixed, white**, and variable in intensity, and are unidirectional showing towards the approach end of the runway.
- Runway centre line lights shall be fixed lights showing **variable white from the threshold to the point 900m from the runway end; alternate red and variable white from 900m to 300m from the runway end; and red from 300m to the runway end**, except that:
 - for runways less than 1,800m in length, the alternate red and variable white lights shall extend from the mid-point of the runway usable for landing to 300m from the runway end.

Runway Touchdown Zone Lights

- Touchdown zone lights shall be provided **in the touchdown zone of a precision approach runway category II or III**.

- Touchdown zone lights shall **extend from the threshold for a longitudinal distance of 900m**, except that, on runways less than 1,800m in length, the system shall be shortened so that it **does not extend beyond the midpoint of the runway**.
- The longitudinal spacing **between pairs of barrettes shall be either 30m or 60m**. A barrette shall be composed of at least three lights with spacing between the lights of not more than 1.5m.
- Touchdown zone lights shall be fixed unidirectional lights showing variable white.



Runway Guard Lights

- A pair of unidirectional flashing yellow lights on both sides of the taxiway before entering the runway
- Unidirectional flashing yellow lights placed at interval of 3m along the taxiway

Stopway Lights

- Stopway lights shall be **fixed, unidirectional lights showing red** in the direction of the runway.

Taxiway Lighting

- Taxiway Lighting system consists of centreline lights, edge lights, guard lights, and stop lights at holding points.
- **Taxiway Edge Lights** - The lights are to be of **fixed variable intensity showing BLUE**, showing all round and up to 30° above the horizontal.

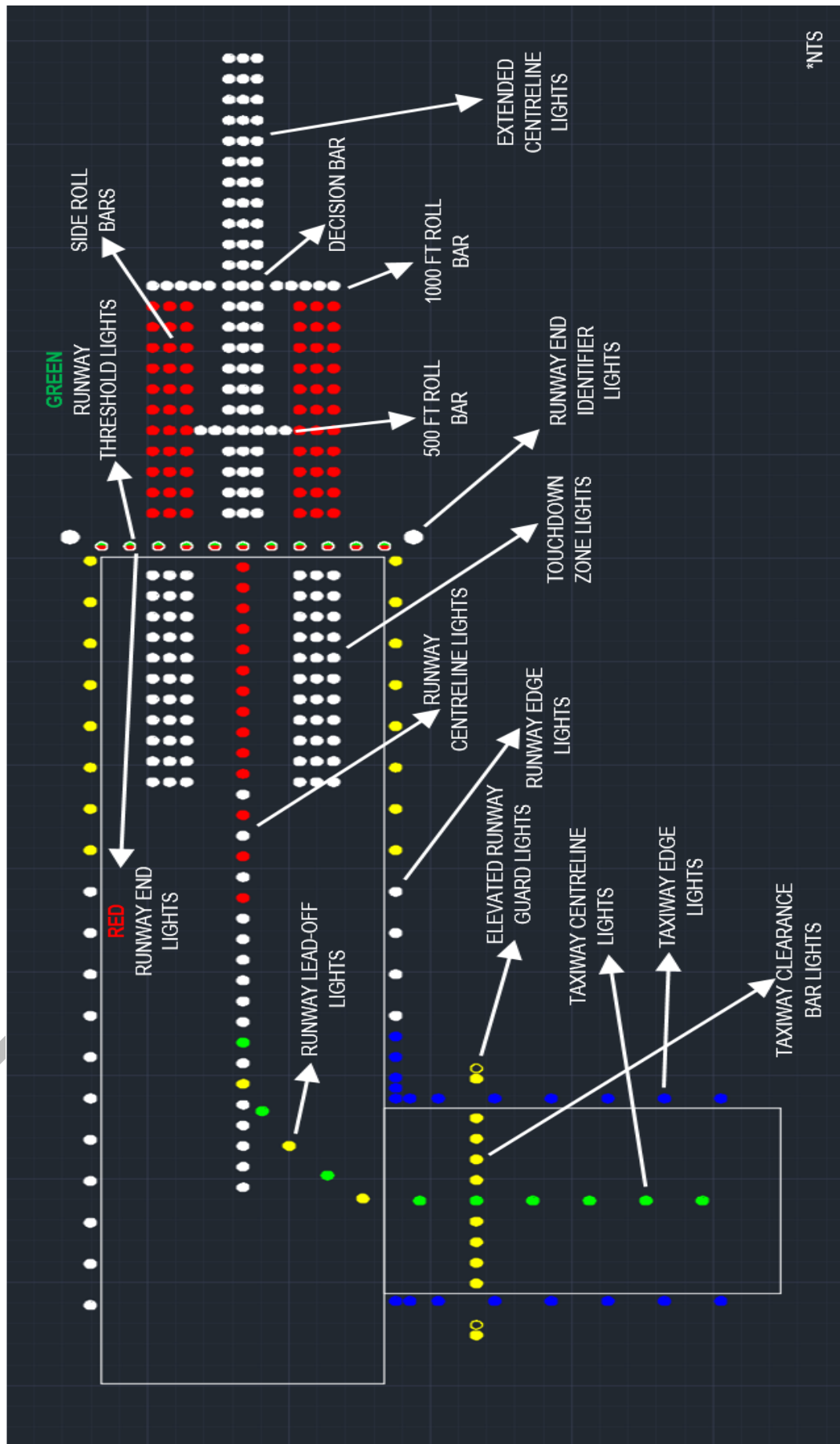
Taxiway Centre Line Lights

- Taxiway centre line lights on a taxiway shall be **fixed light showing green** such that the light is visible only from the aeroplanes on or in the vicinity of the taxiway.
- Taxiway centre line lights on an exit taxiway shall be fixed lights.
- Alternate lights shall show **green and yellow from their beginning near the runway centre line** and thereafter all lights shall show green.

Variable Intensity Lights

The intensity (brilliance) of the following lights is to be variable:

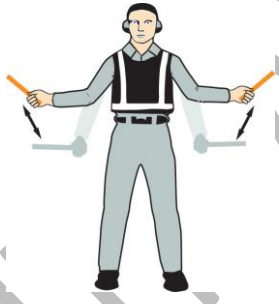
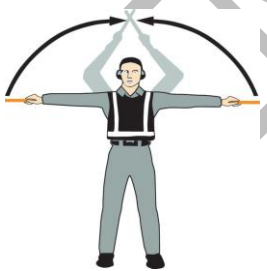
- Approach lighting systems – Simple (420 m), CAT I (900 m), Calvest CAT II & III (900 m)
- Runway edge lights - White, last 600 m yellow
- Runway threshold lights - Steady Green (Uni-directional)
- Runway end lights - Steady Red
- Runway centre line lights - White, Red & White, Red
- Runway touchdown zone lights – White



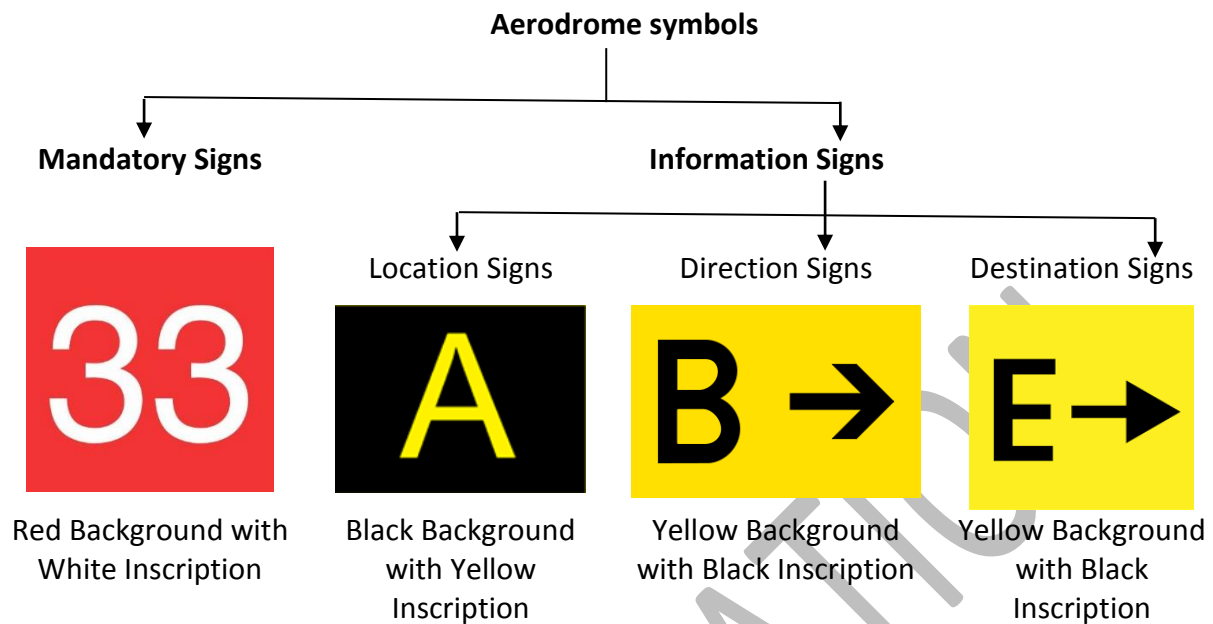
*NTS

Marshall Signals

Signal	Meaning	How to do it
	Turn to your left	Right arm downward, left arm repeatedly moved upward backward. (Speed of arm movement indicating rate of turn.)
	Turn to your right	Left arm downward, right arm repeatedly moved upward backward (Speed of arm movement indicating rate of turn.)
	Move ahead	Arm a little aside, palms facing backward and repeatedly moved upward-backward from shoulder height.
	Chocks inserted	Arms and wands fully extended, palms facing inwards, move arms inwards from extended position.
	Chocks removed	Arms and wands fully extended, palms facing outwards, move arms outwards.

	Start engine(s)	Left hand overhead with appropriate number of fingers extended, to indicate the number of the engine to be started, and circular motion of right hand at head level.
	Cut engines	Either arm and hand level with shoulder, hand across the throat, palm downward. The hand is moved sideways with the arm remaining bent.
		Slow down Arms down with palms toward ground Or Move extended arms up and down from waist to knees.
		Stop (Normal Stop & Emergency Stop) Arms repeatedly crossed above the head (the rapidity of the arm movement should be related to the urgency of the stop ie the faster the movement the quicker the stop).

AERODROME SYMBOLS



- Obstacles above 150m high should be lit by a **steady Red light at night** (Low intensity) and **flashing white light during day** (High intensity).
- All vehicles at airport **have orange or preferably yellow flashing beacon**.
- Apron safety marking can be of any contrasting colour with the stand marking.
- Parking for a **Hijacked plane should be 100m away from normal parking stations**.
- **Aerodrome search and rescue category is based on the overall length and width of the fuselage using the runway.**
- Emergency vehicles on maneuvering area have single conspicuous color preferably red or yellowish green.

HUMAN FACTORS

HYPOXIA is a state of oxygen deficiency in the body sufficient to impair functions of the brain and other organs.

Anaemic Hypoxia

- Anaemic Hypoxia is caused by the **inability of the blood to carry oxygen** and may be due to a medical condition (anaemia) or to carbon monoxide poisoning.

Hypoxic Hypoxia

- The term for the effects of a **shortage of oxygen** is Hypoxic Hypoxia. This can result from a number of reasons. But the most important reason, as far as pilots are concerned, is altitude.

Prevention of hypoxia

For protection, pilots are encouraged to use supplemental oxygen above 10,000 during the day and above 5000 at night.

Dalton's Law

- Dalton Law explains altitude Hypoxia; he explained that the total pressure is equal to sum of partial pressure of gasses which decreases with altitude.
- Chemical Composition is 78% N₂, 21% O₂, 95% Argon, 0.03% CO₂, and rests are rare gasses.
- Oxygen combined with HB in blood is transported by RBC (Red Blood Cells).
- Percentage of O₂ is same at all levels - 21%
- Barometric pressure at 18000 ft is half of barometric pressure at sea level.
- You can survive at any altitude if enough O₂ and heat is provided.
- Breathing 100% O₂ at high altitude (38,000) is equivalent to breathing ambient air at 10,000 ft.
- Breathing 100% O₂ without pressure at 38,000 ft will cause Hypoxia.
- During Hypoxia, set the O₂ value to "Pressure" & pressurized O₂ will be supplied.
- 4 stages of hypoxia are: Indifferent, Compensatory, Disturbance, and Critical.

Symptoms of Hypoxia

- Personality change
- A sense of well-being (Euphoria)
- Impaired Judgment
- Headache
- Tingling in hands & feet
- Hyperventilation
- Muscular Impairment
- Memory & Sensory Loss

- Tunnel Vision
- Impairment of consciousness
- Cyanosis (turning of skin to Bluish color)
- Unconsciousness and ultimately death

Hyperventilation

It is another word for over breathing, Lungs Ventilation in excess of what body needs (Lack of CO₂ in the body) (No proper exchange of O₂ & CO₂ - thus O₂ is not absorbed by Lungs)

❖ Causes of Hyperventilation

- Anxiety
- Shock
- Heat
- G forces
- Motion sickness

❖ Symptoms of Hyperventilation

- Dizziness
- Tingling
- Spasms (Contraction of Muscle)
- Loss of Consciousness
- Increased heart rate
- + CO₂ - blood acidic; -CO₂ - blood alkaline

❖ Treatment of Hyperventilation

- The classic way to treat a patient suffering from Hyperventilation is to make him/her **breathe into a paper bag**.

Hypoxia or Hyperventilation

The natural reaction to shortage of oxygen in the body is to try to obtain more air by breathing faster and deeper. The hypoxic individual may Hyperventilate in an effort to get more oxygen, but this is of little use in an environment of low ambient pressure.

In night it can be difficult to distinguish the symptoms of Hypoxia and Hyperventilation. The appropriate response of pilots must be to assume the worst. **Whenever doubt between Hypoxia and Hyperventilation, assume it as hypoxia and treat accordingly**. Do not assume hyperventilation if it could be hypoxia.

Hyperventilation - after unconsciousness - recovery

Hypoxia - after unconsciousness - death

Carbon Monoxide (CO) Poisoning

- ❖ CO is a colorless, odourless and tasteless gas contained in the exhaust fumes. Also present in the smoke of a cigarette.
- ❖ It can significantly reduce the ability of the blood to carry oxygen and leads to Hypoxia.

- ❖ CO poisoning is likely to occur in aeroplanes where cabin heat is supplied from the exhaust coating.
- ❖ HB has more affinity towards CO; HB combines more rapidly CO as compared to oxygen.
- ❖ Symptoms of CO -
 - Headache
 - Dizziness
 - Nausea
 - Impaired vision
 - Impaired judgment
- ❖ Action to be taken
 - Switch off cabin heat, open cabin ventilation, use fresh air, use O₂ if available
 - Land as soon as possible

Cabin Decompression

- Loss of cabin pressurization can occur in flight.
- Henry explained the release of nitrogen gas bubbles in the body fluids causing Chocks, Creeps and other neurological symptoms in Bends, Knees, Lungs, Brain and under skin.
- Passengers suffering from rapid decompression should wait for 12 hours before next flight.
- Symptoms of decompression sickness can be evident after some time probably when the aircraft is on ground.
- Decompression sickness can occur due to cabin pressure loss when flying at higher altitudes (above 18000 ft).
- After scuba diving pilot may develop symptoms of decompression even when the altitude is less than 18000 ft.
- Flying immediately after scuba diving is prohibited (since it involves risk of decompression sickness without actual decompression).
- Do not fly using compressed air within 12 hours of swimming and avoid flying for 24 hours if a depth of 30 feet has been exceeded during swimming.

TUC (Time of Useful Consciousness)

TUC is the time up to which pilot can act physically and mentally well, after TUC both mental and physical performances degrade.

TUC varies individually and depends on cabin pressure altitude.

Altitude	TUC
18000 Ft	30 Min
22000 Ft	5-10 Min
25000 Ft	3-5 Min
30000 Ft	1-2 Min

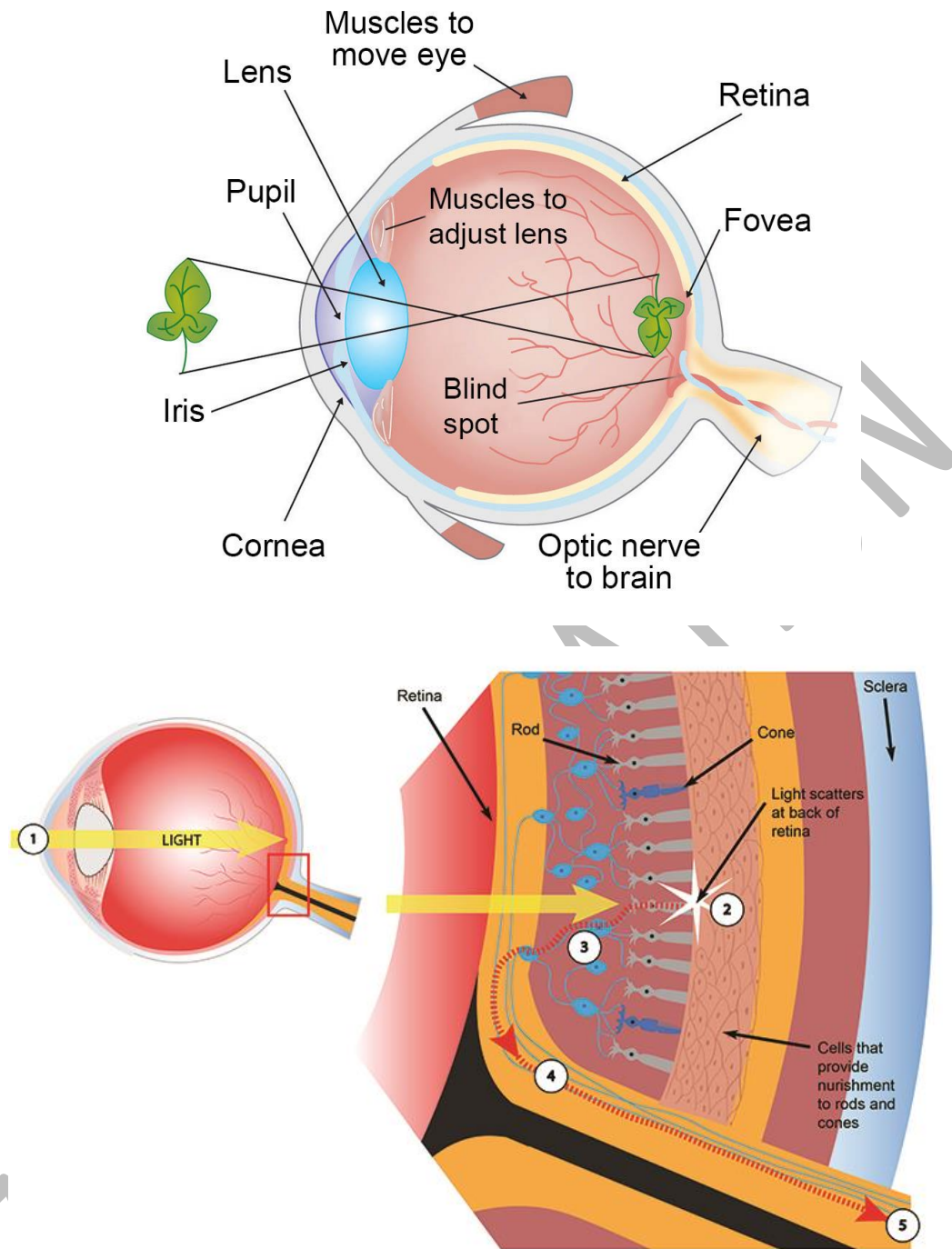
35000 Ft	30-60 Sec (30-90 Sec)
40000 Ft	12 Sec (5-15 Sec)

EYE

- Adaptation - The adjustment of the eyes to high or low levels illuminations. **Bright light: 10 seconds, darkness 7 min for cones, 30 min for rods.**
 - **Cones** are responsible for **day vision** since they are color sensitive.
 - **Rods** are responsible for **night vision**.
 - Fovea - Area in which cones pre-dominate.
 - Fovea Centralis - Best for day vision and no night vision at all.
 - Retina is a light sensitive inner lining containing photo receptors essential for vision.
 - Peripheral vision is important for detecting moving objects.
 - Light enters the eye through the Cornea. The amount of light entering the eye is controlled by cornea.
 - The amount of light entering the retina is controlled by pupil.
 - Night vision scanning: Scanning technique is to look from side to side (15-20 Degree) of an object. (Using off-center view at night)
 - Accommodation of light is done by crystalline lens.
-
- Light Adaptation. When experiencing sudden high levels of illumination the eye quickly adjusts in 10 seconds.
 - It takes time for our eyes to adapt to darkness. This adaptation does take time - about 7 minutes for the cones and 30 minutes for the rods.

The most common factors affecting night vision are:

- (a) Age
- (b) Mild hypoxia
- (c) Cabin altitudes above 8,000 ft
- (d) Smoking
- (e) Alcohol



Visual Defects

Most visual defects are caused by the distorted shape of the eyeball.

Hypermetropia

- In long sightedness, Hypermetropia, the eyeball is shorter than normal and the image forms behind the retina.

Myopia

- In short-sightedness, Myopia, the eyeball is longer than normal and the image forms in front of the retina.

Presbyopia

- Far sightedness linked with age.

In the absence of anything to focus on (Empty field) the natural focus point of the eye is not at infinity, as was long assumed, but at an average distance of between 1 - 1.5 meters.

Cataracts

- Cataracts are nominally associated with the ageing process though some diseases can cause cataracts at any age.
- With time, the lens can become cloudy causing a marked loss of vision.
- The use of contact lenses by aircrew is permitted, under authorized medical supervision.
- Bi-focal contact lenses are prohibited.

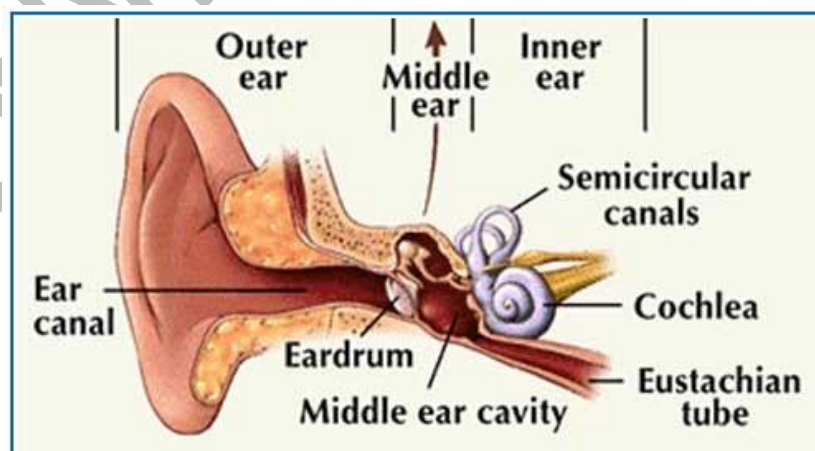
Glaucoma

- Glaucoma is a disease of the eye which causes a pressure rise of the liquid around the Eye.

Color Blindness

- Total colour blindness is a bar to the issue of a flying license. It is caused by a defect in the structure of the colour-sensitive cones in the retina.
- It does not affect sharpness of vision and many people go through their lives with no knowledge that they suffer from this imperfection.
- It is the inability to discriminate colors at varying distances.
- Deficiency of vitamin A causes night blindness, since vitamin A is essential for regeneration of tissues.

EAR



Inner Ear

- Consist of Semi-Circular Canal, Otoliths and Cochlea

Middle Ear

- Consist of tiny Bones, Ossicles

Inner ear is responsible for perception of noise (Cochlea)

- Eustachian tube balances pressures between middle ear and external atmosphere.
- Eustachian tube connects middle ear and Pharynx.
- Vestibular Apparatus (Otoliths + Semicircular Canals) - helps maintains spatial orientation.
- Otoliths senses linear acceleration.
- Semicircular canal senses angular acceleration.
- Intensity of sound is measured in decibels.
- Human Hearing Range: 16 Hz – 20,000 Hz (16 – 20 kHz)
- Prolonged exposure to noise in excess of 90 decibels can end up in noise induced hearing loss.
- **Barotrauma** is likely to cause pain caused by difference in pressure between middle ear and ambient air (either side of ear drum).
 - Solution - Reduced rate of descent or climb or leveling out, gives more time for pressure balancing.
- In case of severe cold, changes in air pressure and blocked Eustachian tube resulting from cold can lead to damage to ear tissues and membranes.

Illusions

1. **Vertigo:** An illusion associated with the vestibular apparatus is Vertigo, **a loss of spatial awareness**, in which the individual experiences a rotating, tumbling or turning sensation (**sudden head Movement creates an illusion that plane has started turning**).
2. **Coriolis Effect:** If a steady turn is being maintained, and then a sudden movement (greater than 30 degree/second) of the head will give an **illusion that the turn rate is changed**.
3. **Samatograval Illusion:** The Samatograval Illusion results from the inability of the semi-circular canals to register accurately a prolonged rotation. Thus a Somatograval Illusion is the **sensation of turning in the opposite direction** that occurs whenever the body undergoes angular deceleration from a condition of sustained angular velocity.
4. **Somatogravic illusion:** The **illusion of pitching up or down as a result of the movement of the Otoliths due to linear acceleration**.

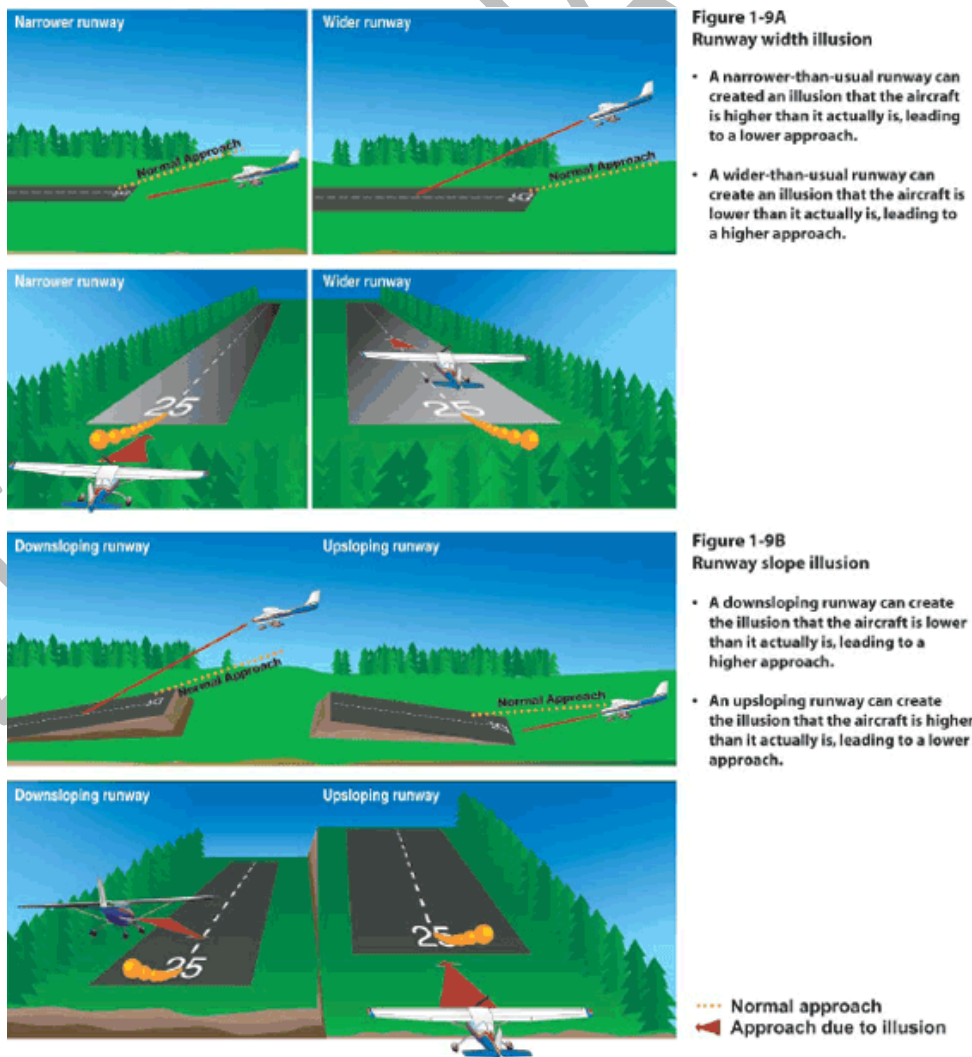
Solution:

 - If IMC: Believe your Instruments
 - If VMC: Look out at the Horizon

5. **Autokinetic Illusion:** (Autokinesis) Illusion in which a **stationary light** if stared for several seconds in the dark without any visual reference **appears to move**.
6. **Narrow Runway:** Pilot may feel he is at a greater height than actual and tendency to land short (Undershoot).
7. **Wide Runway:** Pilot may feel he is lower than actual and has a tendency to overshoot the runway (High round-out)

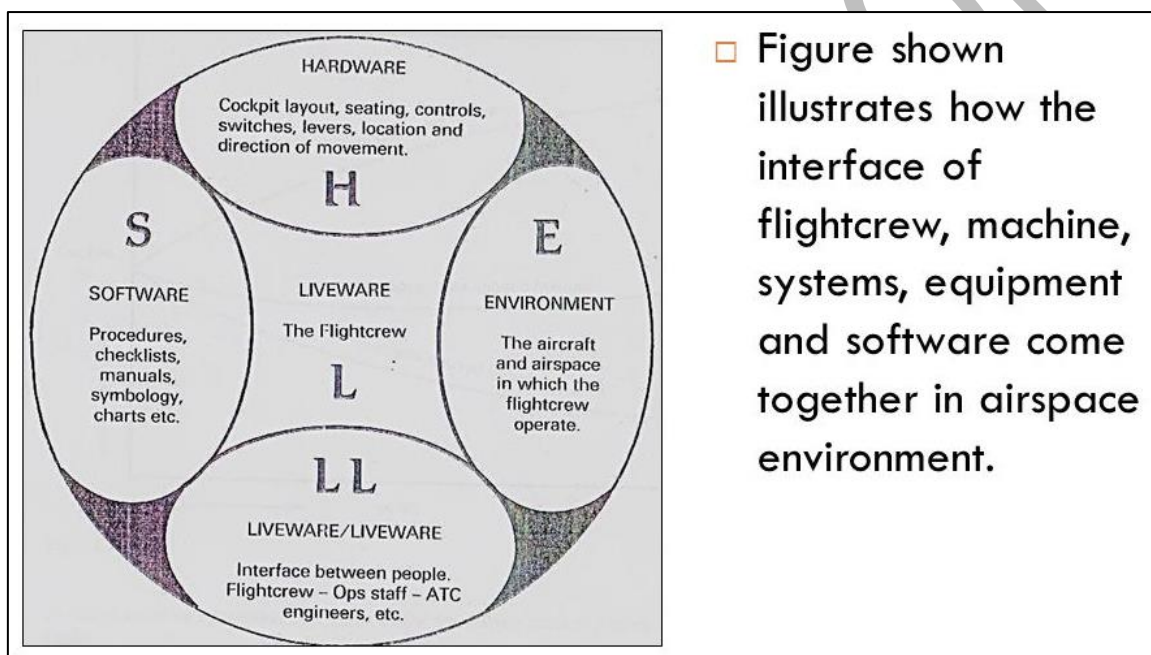
RUNWAY SLOPE

8. **Upslope:** Pilot may feel he is higher than actual; this illusion may cause him to land short.
9. **Downslope:** Pilot may feel he is lower than actual; this illusion may cause him to overshoot the runway (High Round Out).



10. **Black Hole Effect** - In absence of any visual clues on a featureless terrain or darkened surface, pilot feels he is **higher than normal and flew excessively low approaches and crash into terrain short of the runway (tendency to land short).**
11. **Fog or Haze** - May give an illusion of objects being further away than actually is.
12. **Fog gives an illusion of pitch up.**
13. **Seat of Pants** - It is the illusion caused by movement of body parts or muscles or joints. Seat of Pants sense can give false inputs to the body orientation when visual reference is lost.
14. **Centrifugal Force** – it can be interpreted by pilot as rising and falling.

SHEL Model



For example:

- Hardware: 3 pointer Altimeter
- Software: Memory mismatch
- Environment: Bio- logical sleep ware cycle
- Liveware: Eating, drinking habits

Interactions between human beings and other elements of the SHEL model:

1. Liveware-Hardware: humans and machines
2. Liveware-Software: humans and materials
3. Liveware-Liveware: humans and their colleagues
4. Liveware-Environment: humans and the operating environment

Brain has two types of memory:

1. **Short term memory** – Also known as working memory, it can remember about 7 item for about 20 sec.

2. **Long term memory** - It enables us to remember a clearance long enough to write it down.

- The first stage of information process is sensory stimulation.
- Long term memory is an essential component of the pilot knowledge and expertise. It is desirable to pre-activate knowledge stored in long term memory to have it available when required.
- Stress occurs whenever pilot must revise his plan of action and does not immediately have a solution.
- A moderate level of stress may improve performance.
- Stress promotes physical strength rather than mental performance.

Symptoms of overstressed pilot

- Mental blocks, confusion, frustration, rage, deterioration in motor co-ordination, high pitch voice and fast speaking
- Rate of accident in commercial aviation are one accident in Million aircraft movement.

Motor Program – representation of movement that centrally organizes and controls the many degrees of freedom involved in performing an action.

Stages of motor program –

- 1) Cognitive stage: Goal is to develop an overall understanding of the skill
 - 2) Associative stage: Learner demonstrates more refined movement through practice
 - 3) Autonomous/Automatic stage: Motor skills become mostly automatic.
- Grey Out: G force increases upto 3-4g, tunnel vision is induced, oxygen reduced, blood flows upleg
 - Blackout: G force above 5g, individual lose consciousness
 - Red Out: Blood pulls up in brain and in eyes

Respiratory System

- The oxygen required by the body is obtained from the air we breathe. The brain only constitutes approximately 2% of body weight; but it consumes 20% of the total required oxygen for the normal functioning of the body.

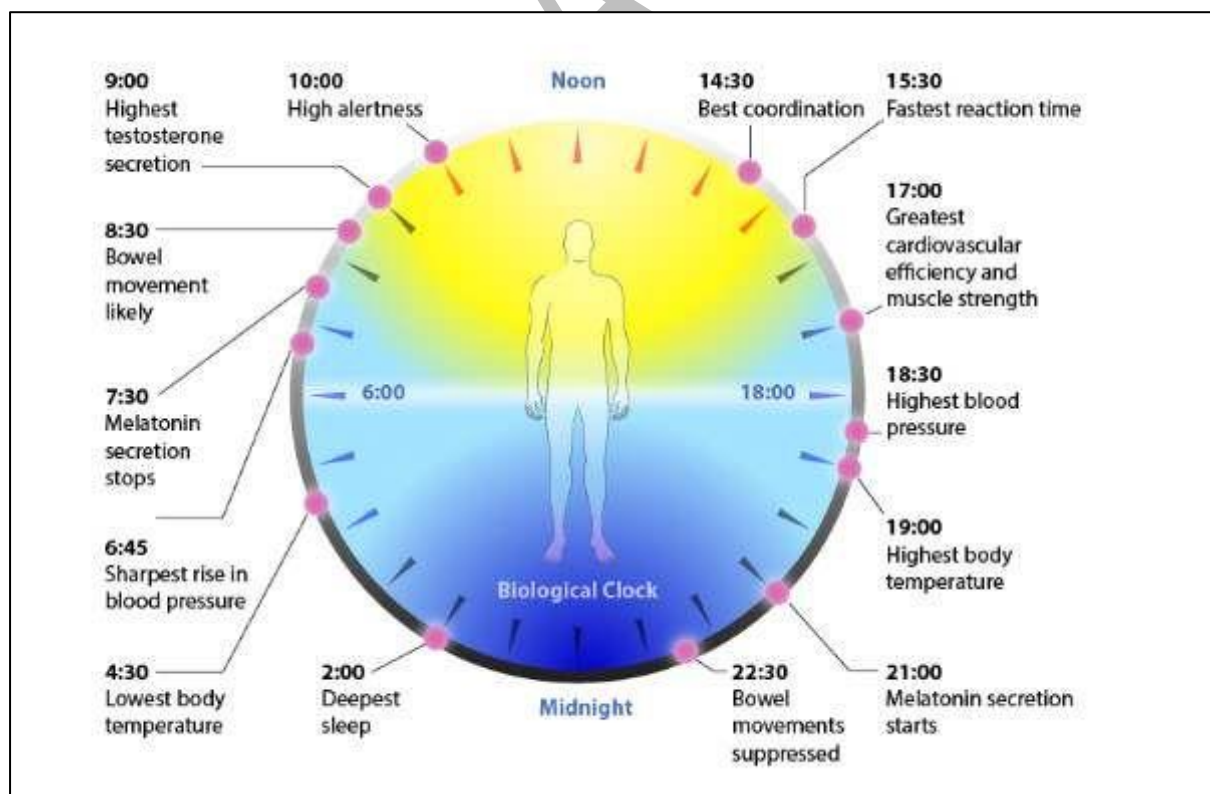
- Tidal Volume is the volume of air inhaled and exhaled with each normal breath. It amounts to about 500 ml in normal male adult.
- Normal rate of breathing of an adult at rest is about 16 - 20 cycles/min.
- The transfer of O₂ from the atvolie to the blood can be described by Law of Diffusion.
- Pulmonary Artery - Oxygen poor & CO₂ rich blood.
- The part of blood without cells is plasma.
- Cardial output - 5 liters/min
- Average pulse of Healthy adult - 60-80 beats/min.

THE CIRCADIAN CIRCLE

The Circadian Circle represents our level of alertness throughout the day. Human performance degradation at circadian lows is one of the major challenges for the aviation industry.

Circadian rhythms generated by biological clock which regulates sleep cycles, body temperature, heart rate, blood pressure, appetite, etc.

As we sleep, our heart rate is lowered and our level of alertness is reduced. Level of alertness is also reduced by our blood pressure which is often lowered after meal times. It is generally accepted that human performance declines at night, when the body and mind desire rest.



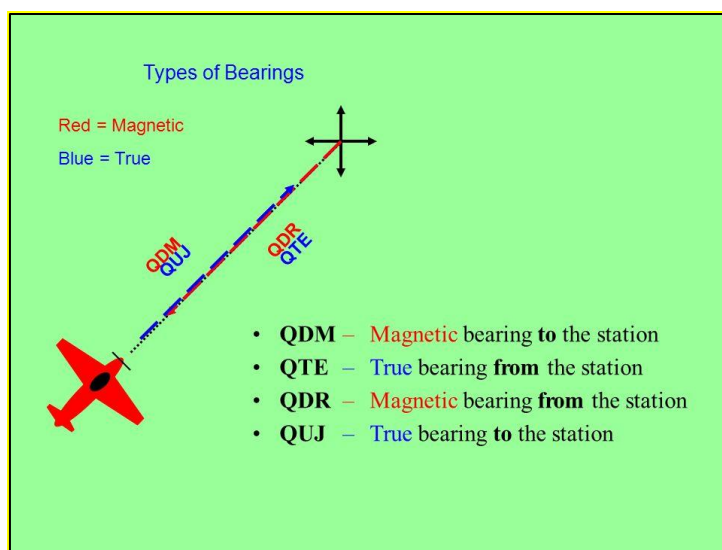
Normal results of an average person during a normal day

Pilots flying across different time zones through the night, are not able to fully apply the Circadian Cycle without understanding or adjusting it. By understanding how our alertness is effected, we can then make adjustments to our diet and sleep to ensure a safe level of alertness in the cockpit.

Following factors should be taken into account by pilots engaged in flights across several time zones:

- **In the absence of all time cues (Free running circadian rhythms), the biological clock has a natural cycle of about 25 hours. With normal cues, the biological clock is reset each day such that it is a synchrony with the solar day. The human circadian rhythm is based on a cycle of about 24 hours.**
- The biological clock adapts slowly with trans-meridian flights. Concerning circadian rhythm disruption (jet lag), the effects of adjustment to destination time may vary greatly between individuals.
- If a stop-over is more than 24 hours, the correct action is to move to the new time as soon as possible.
- In order to minimize the effects of crossing more than 3-4 time zones with a layover more than 24 hours, it is advisable to keep in swing with the rhythm of the departure country for as long as possible and maintain regular living patterns (waking, sleeping alternation and regular meal patterns).
- **The physiological rhythms of a pilot in a new time zone will re-synchronise to this new time zone at a rate of about 1 – 1.5 hours a day.**
- **Adaption after eastbound travel is about 50 percent slower than after westbound flight – adaption time following eastbound travel is about 1.5 days for each time zone change** whereas adaption time following westbound travel is about one day for each time zone change.
- The readjustment of the biological rhythms after a time shift is normally more difficult with flights towards the east.
- The duration of a period of sleep is governed primarily by the point within your circadian rhythm at which you try to sleep.
- Sensorimotor performance is better in the evening whereas intellectual performance is better in the morning.

Communications



QDR - Magnetic bearing (radial) from a facility/station

QDM - Magnetic direction towards facility

QTE - True bearing from a facility

QJ - True bearing to a facility

QNH - Altimeter sub-scale setting to obtain altitude above MSL (**Altitude**)

QFE - Atmospheric pressure at aerodrome elevation (**Height**)

QNE – Height corresponding to pressure setting of 1013.25 HPA (**Flight Level**)

QDL - Series of bearings taken at regular intervals

QFF - Aerodrome pressure reduced to MSL using actual prevailing conditions

S. No.	RT Phrase	Meaning
1.	Affirm	Yes
2.	Correction	An error has been made in this transmission (or message indicated). The correct version is.....
3.	I say again	I repeat for clarity or emphasis
4.	Monitor	Listen out on (frequency)
5.	Back track on runway	Permission to taxi in opposite direction of runway
6.	Runway vacated	No longer occupying an active runway
7.	13500	Transmitted as One three thousand five hundred
8.	118.025	Transmitted as One One Eight Decimal Zero Two Five.

Message in order of Priority -

1. Distress message (Mayday, Mayday)
2. Urgency message (PAN, PAN)

3. Direction finding message
4. Flight safety message

- **Blind Transmission:** (ICAO Definition) A transmission from one station to another in circumstances where two-way communication can't be established but where it is believed that the called station is able to receive the transmission.
- Blind transmission is done twice on the station's frequency.
- Position Reporting:
 - **VFR** flight reports first position after 30 minutes of take-off and thereafter every 60 minutes.
 - **IFR** flight reports its position every 30 minutes after take-off.
- Compulsory position report
 - 1) Call sign
 - 2) Position
 - 3) Time
 - 4) Level
 - 5) Next Position and Time
 - 6) Ensuing Significant Point

(IPTLEE – Identify, Position, Time, FL, estimated next waypoint, Ensuing significant point)

- Readability Scale Meaning

Scale	Meaning
1	Unreadable
2	Readable now and then
3	Readable but with difficulty
4	Readable
5	Perfectly readable

- Test call should not be more than 10 sec.
- Squawk code, frequency, heading, time, runway no – these are communicated as single digits.
- Altitude, height, visibility, runway visual range (RVR) – these are communicated in Hundreds and Thousands.

Cavok: -This is pronounced "CAV-O-Kay" and **means present weather is better than prescribed values**, such that:

- Visibility is 10 km or more
- No cloud below 1500 m (5000 ft) or and no cumulonimbus (CB)
- No significant weather i.e. no precipitation or thunderstorm

SIGMET/AIRMET: en-route weather phenomena which can affect the safety of aircraft operations are issued through SIGMET. It is issued 4 times a day with 4 hours validity by 4 MWO in India.

SIGMET is issued for hazardous in flight weather conditions only such as Severe Turbulence, Volcanic Eruption.

PIREP/AIRREP

It is the report submitted by pilot in-flight to ATC if the pilot experiences hazardous weather or if the weather is different from forecasted weather. It is also known as Pilot report of airborne report.

PIREP/AIRREP is divided into 3 sections:

Section 1: Identity, position, Time, FL

Section 2: ETA/Endurance

Section 3: Winds/Turbulence/Icing/Cloud Burst (Any Hazardous weather)

*Arrangements with Air India exists to submit PIREP/AIRREP.

The Aircraft Act 1934 and Aircraft Rules 1937

This Act was passed by the Indian Legislators and received the assent of the Governor General on 19th August 1934. It is therefore known as the aircraft act 1934.

It extends to whole India and applies -

- A. To all citizens of India where ever they may be**
- B. To the person aboard an Indian registered aircraft wherever they may be.**

The act of 1934 and rules framed in 1937 were brought out in a book which was known as the **Indian aircraft manual**.

The **manual is published every fourth year by the office of DGCA.**

The name of this book has now been changed to **aircraft manual**.

The act of 1934 explains:

1. Prohibition of **slaughtering of animals** and depositing rubbish **within a radius of 10 km** from the ARP.
2. Power to **detain any aircraft by Central Government.**
3. Power to prohibit the **construction of buildings and planting of trees within 20 km** of the ARP. (*The rules for demolition of buildings and trees are given in Aircraft Act 1994).
4. Penalty for flying so close to ground/water so as to cause danger - **2 years of imprisonment or 10 Lakh fine or both.**
5. Removal of Damaged aircraft
 - Aircraft wrecked on water or land may be removed under the **Supervision of Officer of Police or Magistrate or CAD (Civil Aviation Department) officer.**
6. **Landing, Parking and Housing charges** are based on the **total all up weight (AUW)** of an aircraft, contained in the Certificate of Airworthiness.

The Aircraft Rules 1937 contains:

1. **Class Rating**- Single Engine Land, Single Engine Sea, Multi Engine Land and Multi Engine Sea.
2. **Petroleum in Bulk** - Petrol exceeding 900 liters.
3. **Type Rating** - Rating on each aircraft whenever considered necessary by the authority. (B200 - No rating required)
4. **State Aircraft** - Includes military aircraft and aircraft employed for government services, eg- police, customs, and post etc.
5. **Military Aircraft** - Includes Air Force, Army and Naval Aircrafts.
6. **Instrument Time** - Instrument Ground Time + Instrument Flight Time.
7. **ETA (Estimated Time of Arrival)** - The time at which aircraft is estimated to report overhead station or at IAF.

8. **EAT (Expected Approach Time)** - The latest time when the aircraft is expected to leave the hold after completing the instrument approach for final descend for landing.

Aircraft Public Health Rules 1954 contains:

- The **Commander of the aircraft must inform the Health Officer in India at least 2 Hours in advance** on R/T of any person suffering from **quarantinable Diseases**.

Incubation Period:

Disease	Incubation Period
Cholera	Co - 5 Days
Plague	P - 6 Days
Yellow Fever	Y - 6 Days
Relapsing Fever	Re - 8 Days
Small Pox	S - 14 Days
Typhus	T - 14 Days

Carriage of Dead bodies and cremated remains

- Dead body or human remains of those who died because of the following diseases cannot be brought to India.
 - ❖ **PAYG - Plague, Anthrax, Yellow Fever, Glanders**
- But cremated ashes of dead bodies or human remain of those died because of PAYG diseases can be brought to India.
- **Cremated ashes shall be placed in an Urn or casket** having an outer packing of suitable material.
- In case of **uncremated remains** - It should be **hermetically sealed in a zinc container**.

Air Defence Clearance

- **ADC is to issued expect when flying within a radius of 5 NM from the ARP and vertical limit of 1000 ft** (Within 5 NM and below 1000ft - no ADC is required)
- ADC is **valid for the entire route** irrespective of number of halts in between.
- If there is **more than 30 min delay in departure, then new ADC** is to be requested.
- All aircrafts need **ADC 10 min before entering ADIZ** (Air Defense Identification Zone).

Carriage of Cabin Crew

No. of Seats (excluding Flying Crew)	No. of Cabin Crew Required
10 - 49	1
50 - 99	2
100 - 149	3

150 - 199	4
200 - 249	5
250 - 299	6

- **Carriage of arms and explosive (Dangerous Goods):** Only with the **written permission from the Central Government**
- **Carriage of Mails:** Only with the **permission of DGPT** (Director General of Post and Telegraph)
- **Carriage of Prisoners:** With **permission of DGCA, DDG (Deputy Director General), DRI (Director of Regulation and Information), CAD (Civil Aviation Department)**
- **Carriage of person suffering from mental disorders or from epilepsy:** Only when accompanied by the medical practitioner & medicines
- **Photography at an aerodrome** can be done with the **permission of DGCA.**
- **Operational manual is prepared by the operator (Indigo) and approved by DGCA.**

Refuelling

The following distances are to be maintained during the refueling process:

- **Building - 15 M**
- **Person - 15 M**
- **Smoking - 30 M (Naked Flame)**
- **Jet Blast - 43 M**

*For safety reasons, a person shall remain at least 200ft away from jet engine.

Maximum speed allowed for civil aircraft is 0.92 Mach

FDR (Flight Data Recorder)

- It is placed at rear of the aircraft in the tail section and helps to know aircraft performance and investigation.
- It starts automatically once the aircraft moves under its own power.
- It records flight data for 25 hours.
- Maximum 1 hour of oldest data may be erased for testing FDR.
- Color of FDR is orange.

CVR (Cockpit Voice Recorder)

- Records **30 Min to 2 Hours of crew conversation** to help in investigation. (CVR in Indigo- 12 hrs).

Documents to be carried on Board:

1. Certificate of Registration
2. Certificate of Airworthiness
3. Crew Flight License
4. Radio License

5. Passenger List
6. Cargo Manifest
7. Logbook of Aircraft
8. Emergency Check List
9. OPS Manual (Operations Manual)
10. Route Guides
11. Medical License

- ❖ **Certificate of Registration is valid till the time aircraft is destroyed in an accident.**
- ❖ **Smoking** is permitted or prohibited as **written in the Certificate of Airworthiness.**
- ❖ **Crew log books** should be **preserved for 5 years** from the date of last Entry
- ❖ **Dropping of paper leafs/flower** from air is done with the **permission of local District Magistrate.**
- ❖ **Succession of Command is given by Operator**

Oxygen Requirements

Altitude	Oxygen Requirement
Upto 10000 ft	No Oxygen needed
Flying between 10000 to 13000 ft for more than 30 mins	Crew members + 10% of passengers
Above 13000 ft	Crew members + 30% of passengers
Above 15000 ft	Crew members + All passengers

Above 49000ft the cosmic radiation received by each crew member or passenger can be harmful and needs to be monitored for crew members.

Fuel Requirements

Piston Prop Aircraft	Turbojet Aircraft (Turbine Engine)
When ALTN Aerodrome is required (IFR Flight) • Fuel to DSTN + FUEL to ALTN + 45 min of holding fuel	When ALTN Aerodrome is required (IFR Flight) • Fuel to DSTN + FUEL to ALTN + 30 min of holding fuel over alternate Aerodrome @ 1500 ft AGL
When ALTN is not required (VFR Flight) • Fuel To DSTN + 45 min of holding fuel	When ALTN is not required (VFR Flight) • Fuel to DSTN + 30 min of holding fuel @ 1500 ft AGL
When ALTN is not available (Isolated Aerodrome) • Fuel to DSTN + 45 min of fuel and	When ALTN is not available (Isolated Aerodrome) • Fuel to DSTN + 2 hours of fuel at

15% of trip fuel or 2 hours of fuel at normal consumption (whichever is less)	normal Consumption
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Hydro planning / Aqua planning

Hydro planning is a condition when aircraft tyres are rotating over a thin film of water. If flight manual doesn't contain the landing distance required on a wet runway then pilot must add 15% extra of the distance calculated on a dry runway.

Damp Runway

A runway is considered damp when the **surface is not dry**, but when the moisture on it **does not give a shiny appearance**.

Contaminated Runway

A runway is said to be contaminated if more than **25% of the surface area is covered by water more than 3 mm deep** or by slush or loose snow equivalent to 3 mm of water.

Wet Runway

A runway is considered wet when the **runway is covered with water**, or equivalent, less than specified in contaminated runway (Less than 3 mm), which causes **runway surface to appear reflective**, but without significant areas of standing water.

- Mu-meter measures co-efficient of friction for every one third part of the runway.

Friction Co-efficient	Estimated Surface Friction	Code
0.4 and above	Good	5
0.39 - 0.36	Medium to Good	4
0.35 - 0.3	Medium	3
0.29 - 0.26	Medium to Poor	2
0.25 and below	Poor	1

Flight Duty Time Limitations (FDTL)

Domestic Flights:

	Flt Time	Flt Duty Time	No. of Landings
2 Crew	8	11	6
Day	9	12	3
Night	9	12	2

International Flights:

	Flt Time	Flt Duty Time	No. of Landings
2 Crew	10	13	1
Day	9	12	3

Night	9	12	2
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Max flying hours allowed in a:-

- Day - 8 hrs
- Week - 35 hrs
- Month - 125 hrs PIC or 100 hrs PIC + 30 hrs Co-pilot time

FDTL Extension:

	Max. in a day	Max. in 30 consecutive days
FLT Time	1.5 hrs	3 hrs
FLT Duty Time	3 hrs	6 hrs

No. of First Aid Kits required:

No. of Passengers	No. of First Aid Kits
0-100	1
100-200	2
200-300	3
300-400	4

No. of Fire Extinguishers required:

No. of Passengers	No. of Fire Extinguishers
7-30	1
31-60	2
61-200	3
201-300	4
301-400	5

Annexure – 9

Facilitation

General declaration	Cargo Manifest	Stores List
Aircraft registration, Flight No, Place of Departure and Arrival, No. of Crew and passengers	Airway Bill No, No. of package, nature of good, Shipper details	List of goods in aircraft cabin, No. of bottles etc.

- 2 or 3 copies of each is to be presented at time of departure.
- All these documents are accepted in handwritten Block lettering in ink.
- An aircraft is admitted temporarily free of custom duty for a period decided by the state.
- Passengers in the contracting state due to lack of any connecting flights are allowed to remain in the state for the period of 2 days.
- Flights landing in international airports without informing must give at least 2 hours advance notification on their arrival.
- Contracting state shall accept oral declaration of baggage from the crew members.
- The crew doesn't need a passport with CMC (Crew Member Certificate).

Annexure – 12

Search and Rescue

The Color Identification of droppable containers:

1. **Red for Medical supplies and first AID equipment**
2. **Blue for food and water**
3. **Yellow is for blanket and protective clothing**
4. **Black for miscellaneous equipment**

Ground-Air Visual Signals

Signal	Meaning
V	Require Assistance
X	Require Medical Assistance
SOS	Save Our Souls
Y	Yes or Affirmative
N	No or Negative
NN	Nothing found, Will continue to search
LL	Found all personnel
LLL	Operation completed
XX	Not able to continue, Returning to base

Annexure – 13

Aircraft Accident and Incident Investigation

Accident: An accident is defined as

- 1) A person is fatally or seriously injured as result of
 - Being in the aircraft or direct contact with any part of the aircraft
 - Directly exposed to jet blast
 - Serious injury is considered as an accident.
 - An injury which is experienced by a person is an accident and which
 - Requires hospitalization for more than 48 hours
 - Result in a fracture of a bone or 3-degree burns.
 - The aircraft sustained damage would normally require major repair or replacement of the affected component.
 - Accident must be reported to DGCA within 24 hrs.
- Example: Landing in a wheat field and nose wheel is damaged.

Incident: An occurrence, other than accident, which affect the safety of the operation.

Serious incident is an incident involving circumstance indicating that almost an accident has occurred.

Example: Aborted take-off, aircraft on collision course, aircraft on stopway.

- The state in which accident took place is responsible for investigation of accident.
- Purpose of investigation is to prevent future accidents and help judges.
- Report of the accident is to be submitted to all state of registry, state of operator, state of design and manufacturing, and also to ICAO if aircraft weight is greater than 2250 kg.
- Report must be submitted in one of the working languages of ICAO.
- State of accident occurrence may require state of design or manufacture to participate in investigation if the All Up Weight (AUW) of the aircraft is over 100000 kg.
- **AIRPROX** (Aircraft Proximity) – AIRPROX is a situation in which, in the opinion of a pilot or ATS personnel, the distance between the aircrafts as well as their relative positions and speed have been such that the safety of the aircrafts involved may have been compromised.

Annexure – 18

Safe Transport of Dangerous Goods by Air

- Dangerous goods are substances which could cause hazard to property or health and thus carried by air.
- ICAO Doc 9284 entitled as “Technical instruction for safe Transport of Dangerous goods by air”, which must be completed before carrying dangerous goods.
- Doc for dangerous goods or hazardous materials should be prepared by Shipper (The company which is shipping goods).
- For Captain to decide whether a particular item/goods can be transported in the aircraft or not, he should refer to the Doc 9284.

Minimum Equipment List (MEL)

- **Master minimum equipment list (MMEL)** is a document prepared by the manufacturer and attached in the flight manual which contains list of items/systems which must be operative for dispatch of the aircraft.
- **MMEL** is published and established by manufacturer and approved by DGCA.
- The MEL will consider each individual operator's particular aircraft equipment and operational conditions.
- **So MEL** is published by the operator in the operator manual and approved by DGCA.

Ground De-icing

- Icing on an aircraft has following disadvantages:
 - Weight increases, stall speed increases, stall angle decreases
- Thus, before flight Captain must ensure that the external surface of the aircraft is free from any ice formation which may degrade aircraft performance except within the limits allowed as specified in the manual.
- Anti-icing device prevents the formation of ice on the aircraft surface and is a continuous process.
- De-icing devices removes or breaks ice when formed. Fluid spray is a de-icing technique.

Bird Strike

- Hazardous bird concentrations are present during migration periods in large temporal and spatial scales but can occur occasionally throughout the year at regional scale.
- The most effective way of scaring birds is shell crackers.
- 90% of the bird strikes occur under 500 m.
- Birds fly away about 2 seconds beforehand ahead of the aircraft taking off, when aircraft is at an average speed of 135 kts.
- If you encounter birds during take-off or climbing or cruise, immediately inform ATC to avoid other aircraft on that route.

Fire/Smoke

- **Class A Fire:** Organic material – Wood, Paper, cotton etc. Use - H_2O , CO_2 , Dry Chemical, Halon, Water/glycol
- **Class B Fire:** Liquid material – Fuel fire. Use - CO_2 , Halon, Dry chemical
- **Class C Fire:** Liquid Gaseous Fire, electrical equipment. Use - CO_2 , Dry chemical and Halon
- **Class D Fire:** Metal Fire. Use - Dry powder, sand.
- **Flight Deck Fire:** Use Halon and BCF (Bromochlorodifluoromethane).
- **Brake Fire:** Dry powder and sand can be used.
- CO_2 is good for paper fire, plastic fire, fuel, electric fire but it should not be used for metal fire or brake fire since it causes explosion. (Class A, B, C)
- **Water** is used on combustible material fire, should not be used for electrical, fuel or brake fires. (Class A)
- **Dry Chemical (Dry Powder):** best known for wheel and brake fires. Used for fuels, wood, plastic, fabric, paper (Class A, B, D).
- **Halon** is 3 times as effective as CO_2 extinguisher and can be used in case of almost all fires, except metal fires (Never use halon in metal fires).

Noise Abatement Procedures

Departure Climb:

- For the departure climb it is ensured that the safety of flight operations is maintained while minimizing exposure to noise on the ground.
- Noise abatement procedures shall **not be executed below a height of 240 m (800 ft)** above aerodrome elevation.
- The noise abatement procedure specified by an operator for any one aeroplane type should be the same for all aerodromes.
- Noise abatement climb procedures are **not to be used in conditions where wind shear warnings exist, or the presence of wind shear or downburst activity is suspected.**
- Maximum acceptable body angle specified for an aeroplane type shall not be exceeded.

During Approach

- Will not involve use of reverse thrust
- **Aeroplane should be in the final landing configuration after passing the outer marker or at 5 NM from threshold runway.**
- Noise abatement shall not be the determining factor in use of a runway
 - a. When cross wind component, including gust, exceeds 15kt
 - b. When the tailwind component, including gust, exceeds 5 kt
 - c. When the runway is not clear or dry
- No turns will be made coincident with a reduction of power
- No ILS or visual guidance decides a runway being used for noise abatement procedures if landing in VMC